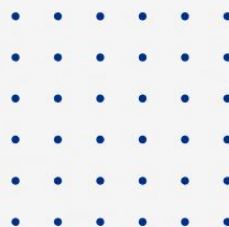
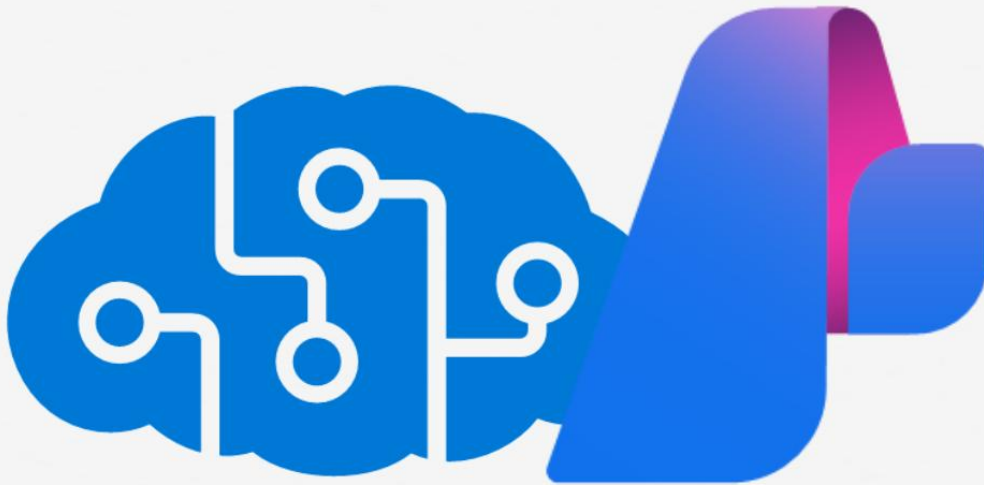


Azure AI/Gen AI Certification Training Lab Guide



System Requirements: Lab Guide

Our comprehensive lab guides ensure practical learning experiences that enhance retention and understanding. 96% of our users report better comprehension through hands-on practice.

The recommended hardware and software requirements for our training are given below:

Hardware Requirements

- **Processor:** 64-bit processor with 2.5 GHz or faster; Intel i5 or AMD Ryzen 5 or higher.
- **Memory:** 8 GB or 16 GB RAM recommended for smooth multitasking.
- **Storage:** At least 256 GB SSD recommended for faster boot times and data access.
- **Display:** Full HD display with a resolution of 1920 x 1080 or higher recommended

Software Requirements

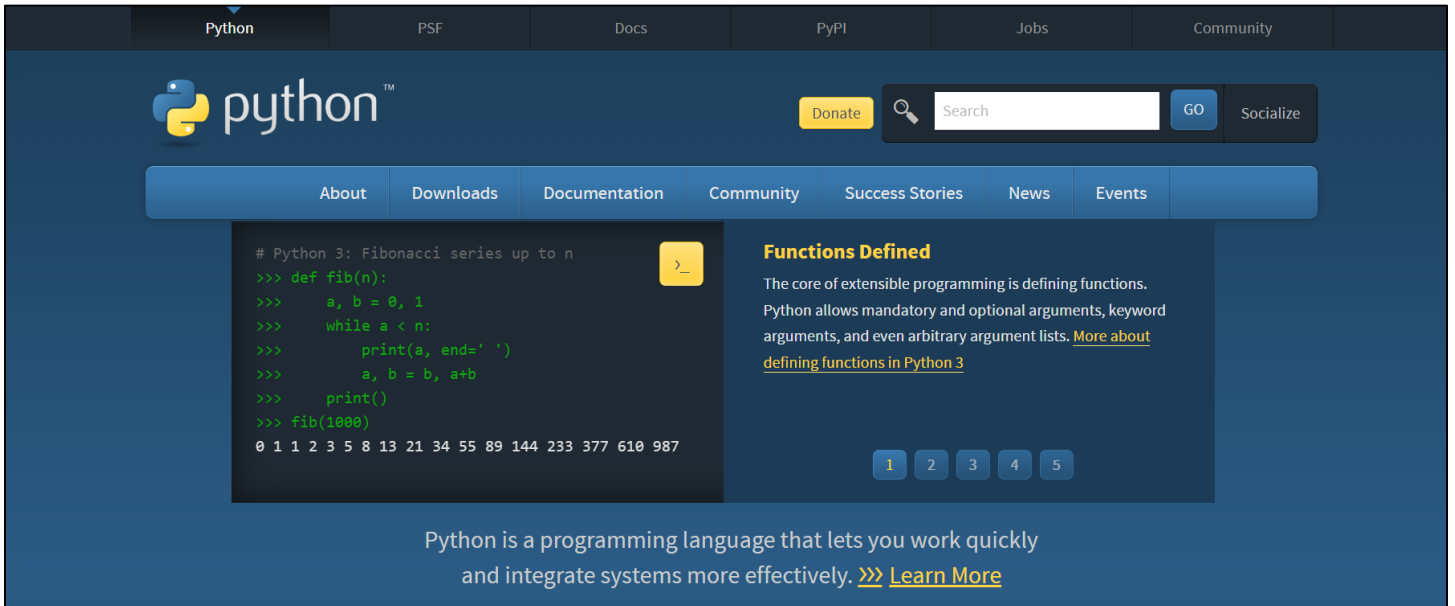
- Supported OS: Windows 10, Windows 11
- Not Supported OS: Windows 7, Windows 8, Windows 8.1
- VS Code IDE
- PyCharm IDE
- Node.js 20 or Higher
- Python SDK 3.10 or higher
- Git
- ScholarHat AI Cloud Sandbox

Software Download Links

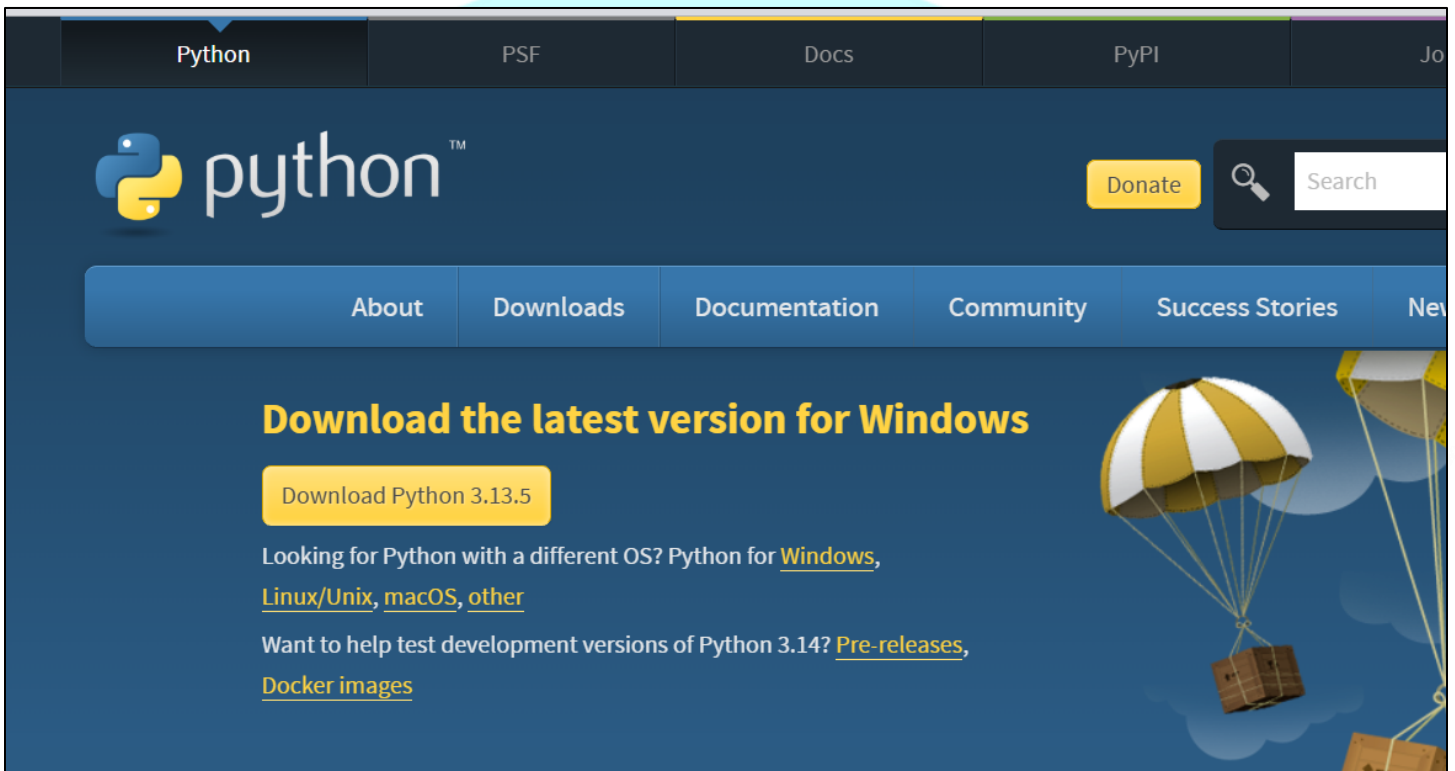
- Python SDK: <https://www.python.org/downloads/>
- Visual Studio Code: <https://code.visualstudio.com/>
- Git: <https://git-scm.com/downloads>
- PyCharm: <https://www.jetbrains.com/pycharm/download/?section=windows>
- Node.js: <https://nodejs.org/en>

1. Python SDK

- **Step 1: Download Python from <https://www.python.org>.**



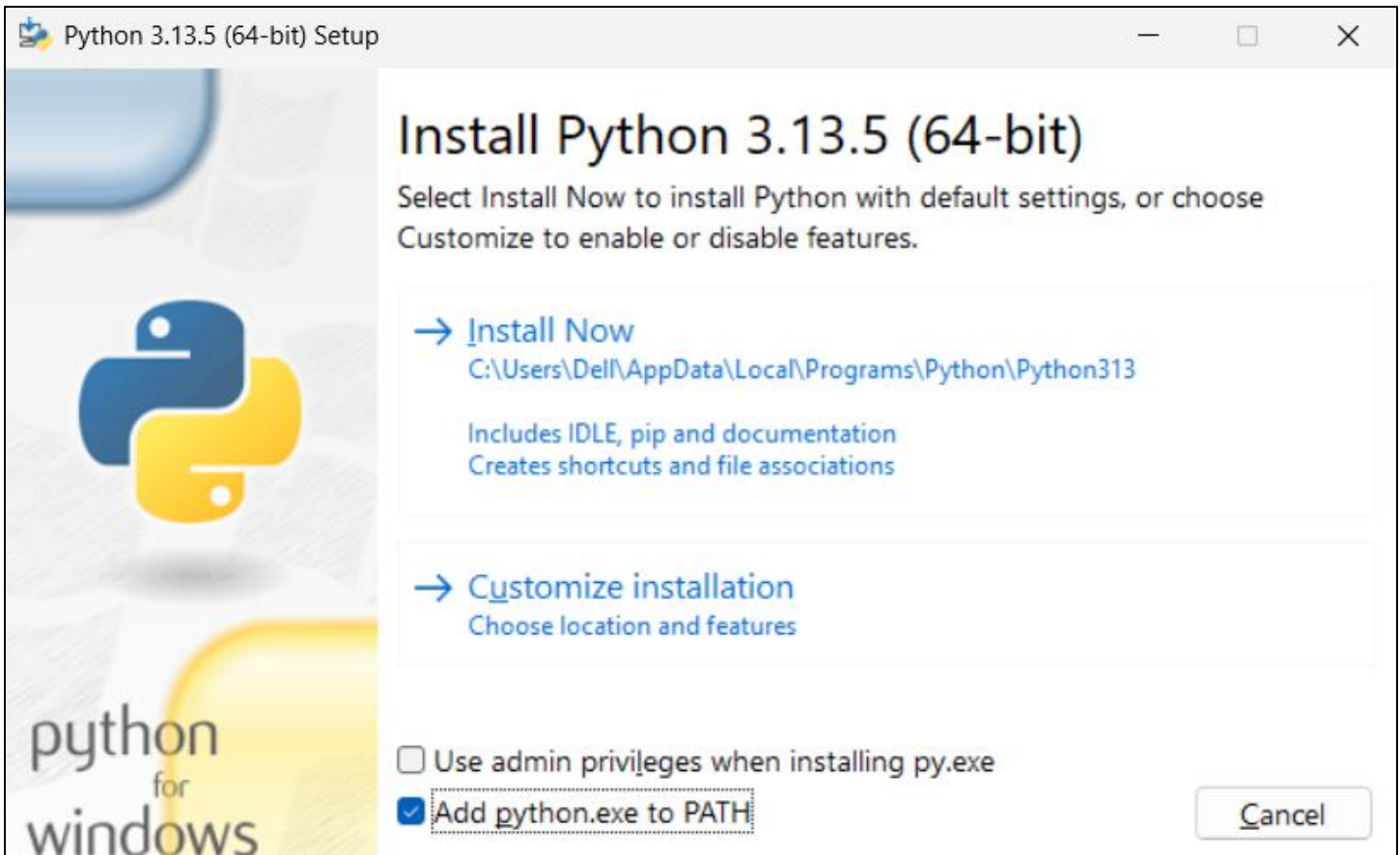
- **Step 2: Select Version.**



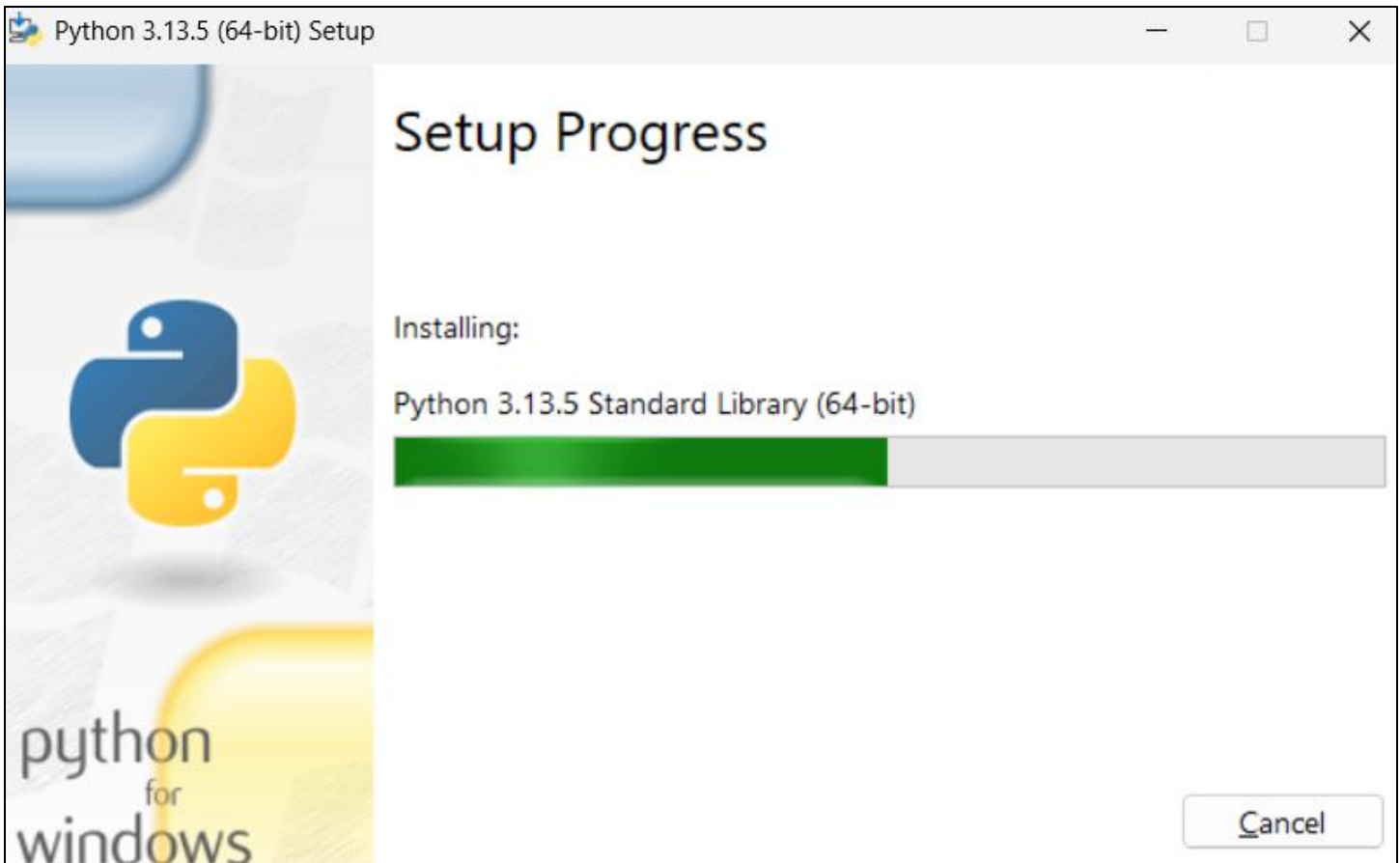
- **Step 3: Run the downloaded installer**



- **Step 4: Check "Add Python to PATH" and click "Install Now".**



- **Step 5: Let the installation complete.**



- **Step 6: Open Command Prompt/Terminal and type `python --version` to verify.**

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

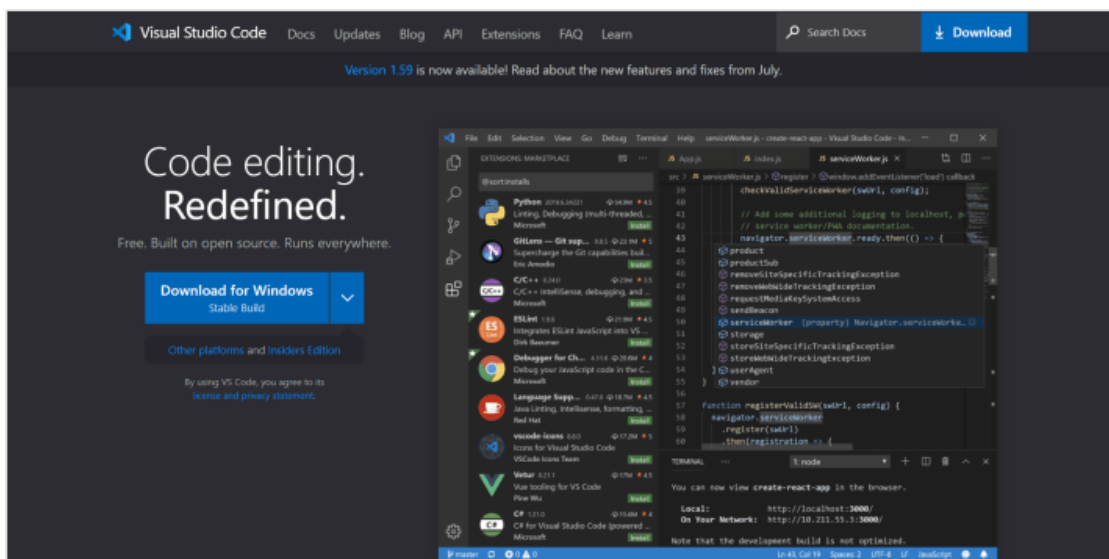
Warning: PowerShell detected that you might be using a screen reader and has disabled PSReadLine for compatibility purposes. If you want to re-enable it, run 'Import-Module PSReadLine'.

PS C:\Users\Dell> python --version
Python 3.13.5
PS C:\Users\Dell> |
```

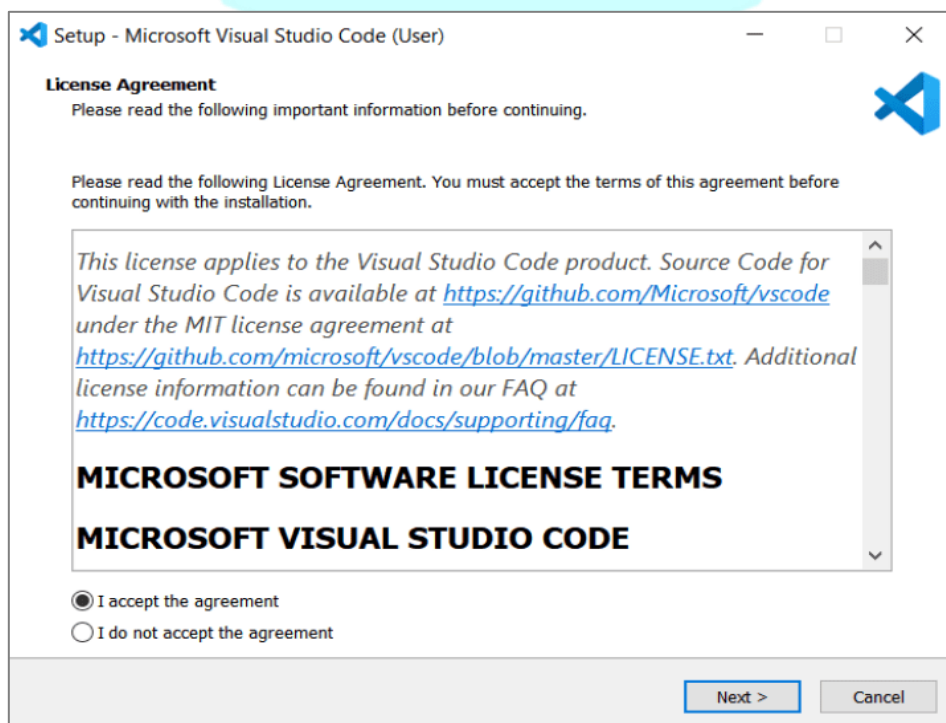
2. Visual Studio Code Installation

VS Code is an open-source code editor developed by Microsoft for Windows, Linux, and Mac OS platforms. Here we discuss the steps to install Visual Studio Code or VS code on Windows. It has built-in support for TypeScript, JavaScript, Node.js, and almost all programming language/library/framework-based plugins are available in VS studio right now.

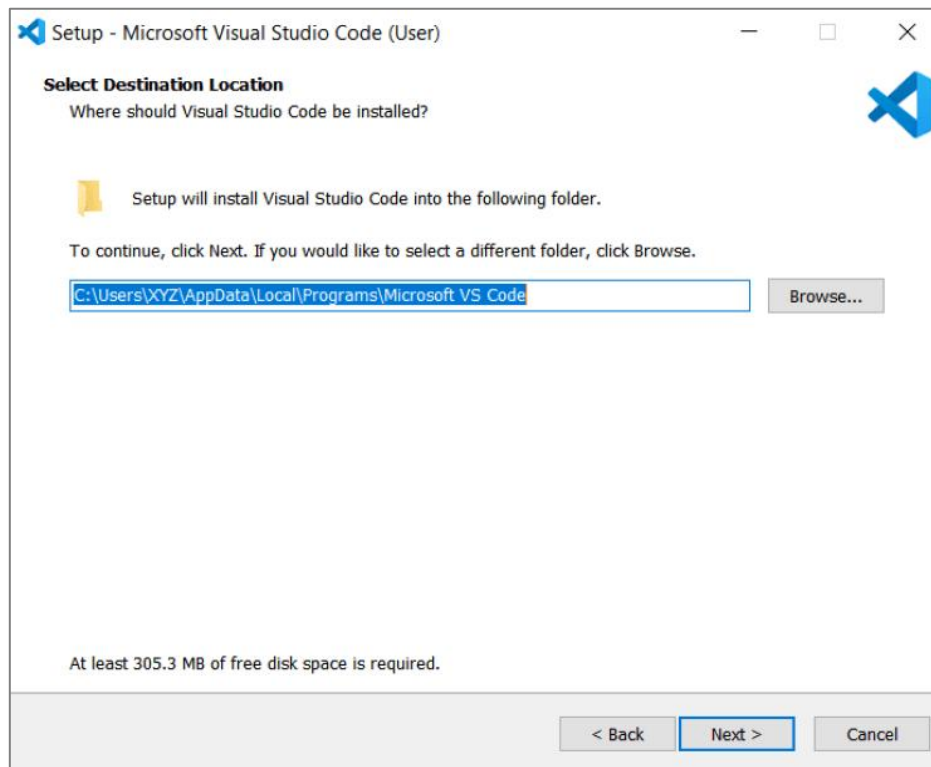
- Step 1: Go to <https://code.visualstudio.com/> and download the Setup file according to your Operating System.



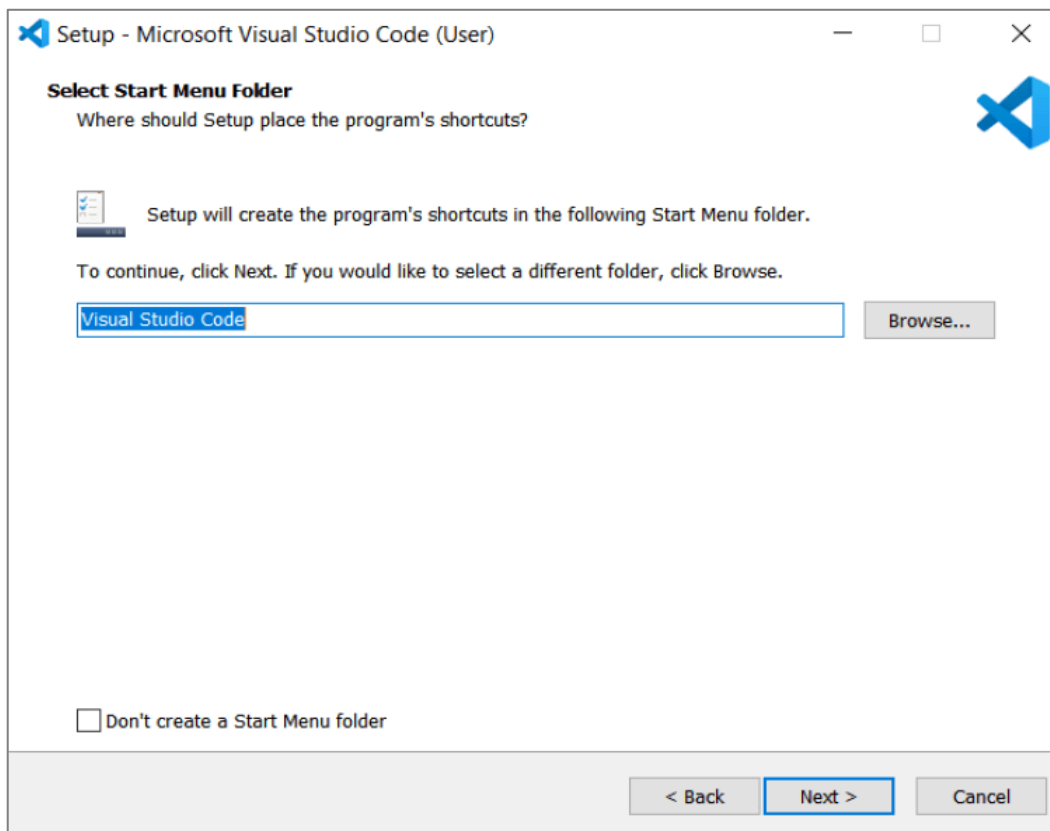
- Step 2: After Downloading the file from the link, open the installer.
- Step 3: Accept the license agreement and click on Next.



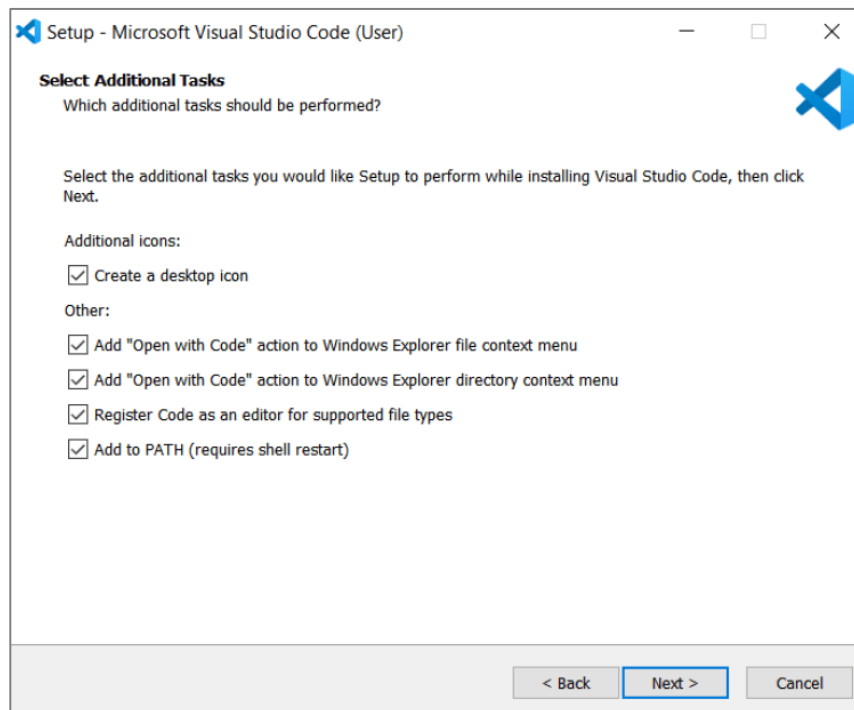
- **Step 4: You can Select the destination where you need to install, by default it will be installed in the folder given in the below image.**



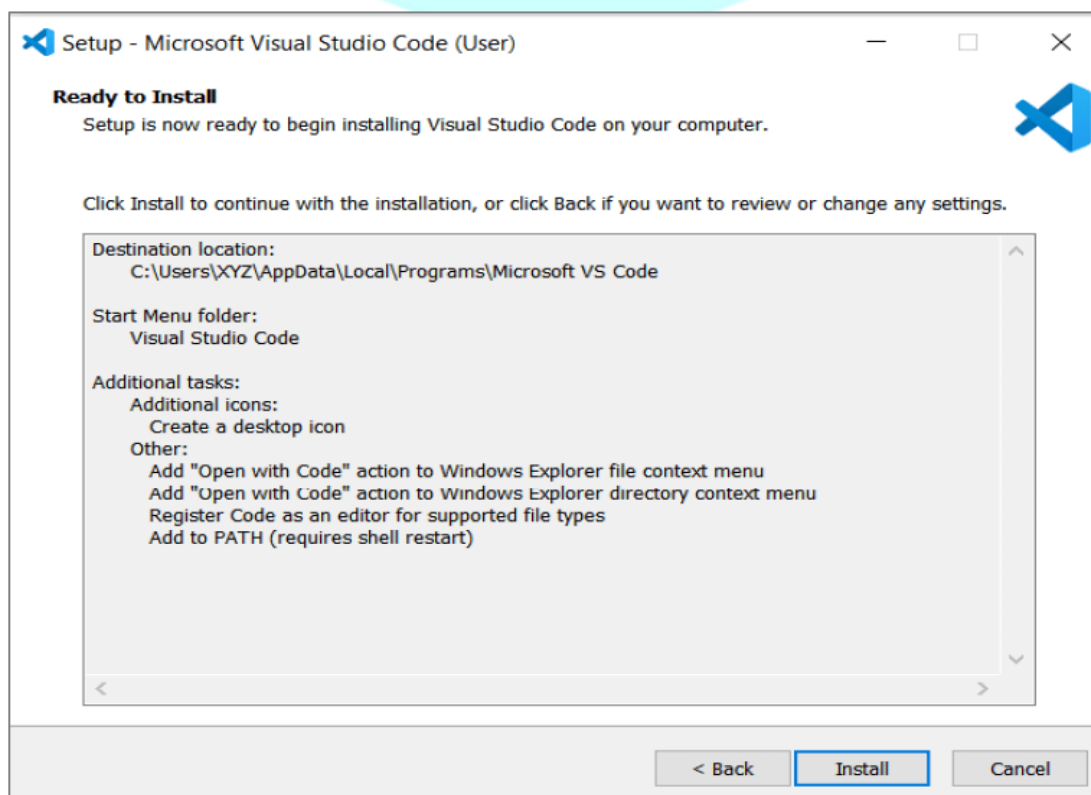
- **Step 5: Click on Next and It will create folder in start menu.**



- **Step 6: Click all the checkboxes and it will be helpful and it will also add to the PATH.**



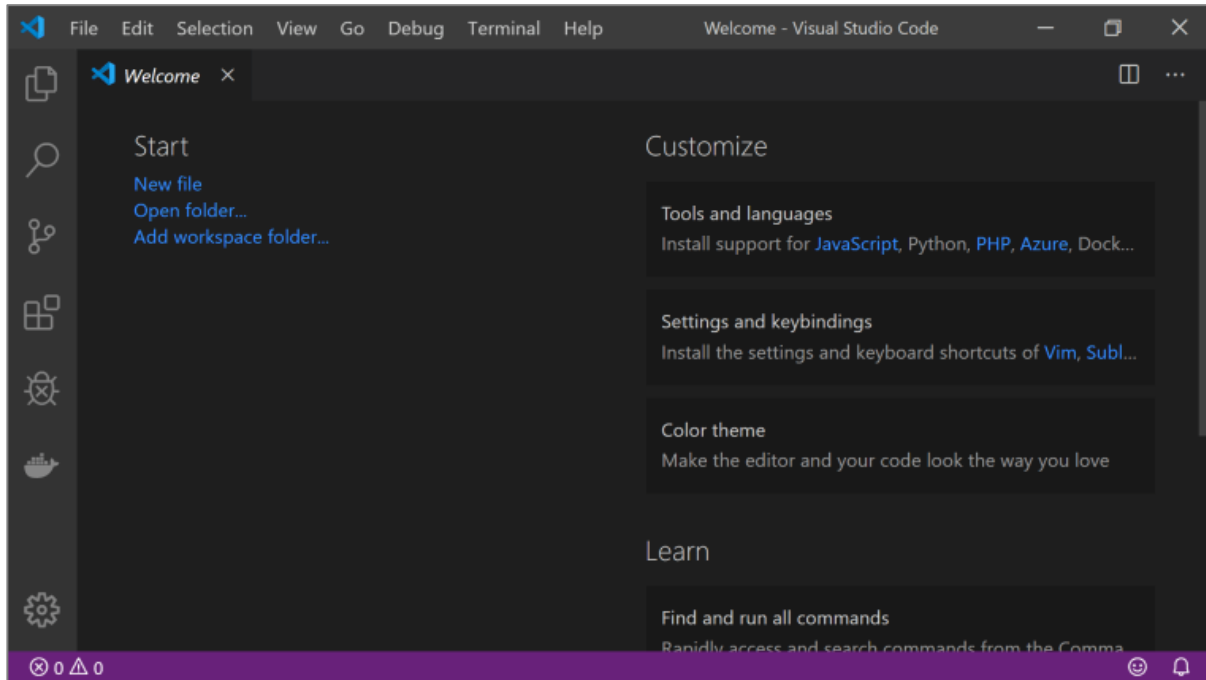
- **Step 7: Review Settings**



Now it will show all your settings there to install VS Code.

- **Step 8: Launch VS Code**

Then you will see the Welcome Screen of VS Code as shown below.



Visual Studio Code has successfully installed.

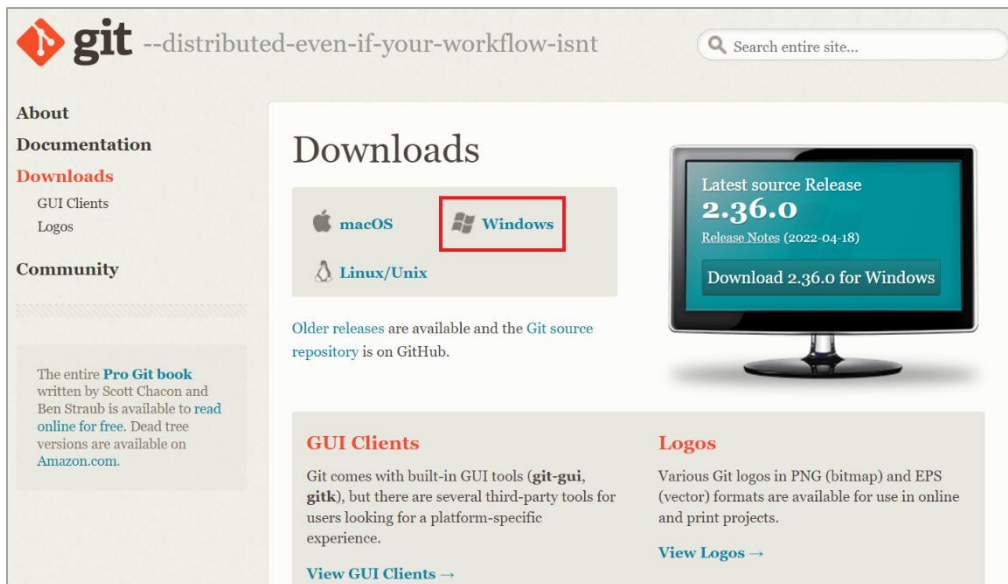
3. Git Installation

Git is a widely used open-source software tracking application used to track projects across different teams and revision levels.

Installing Git prompts you to select a text editor. If you don't have one, we strongly advise you to install it prior to installing Git.

- **Step 1: Download to the official website- <https://git-scm.com/downloads>**

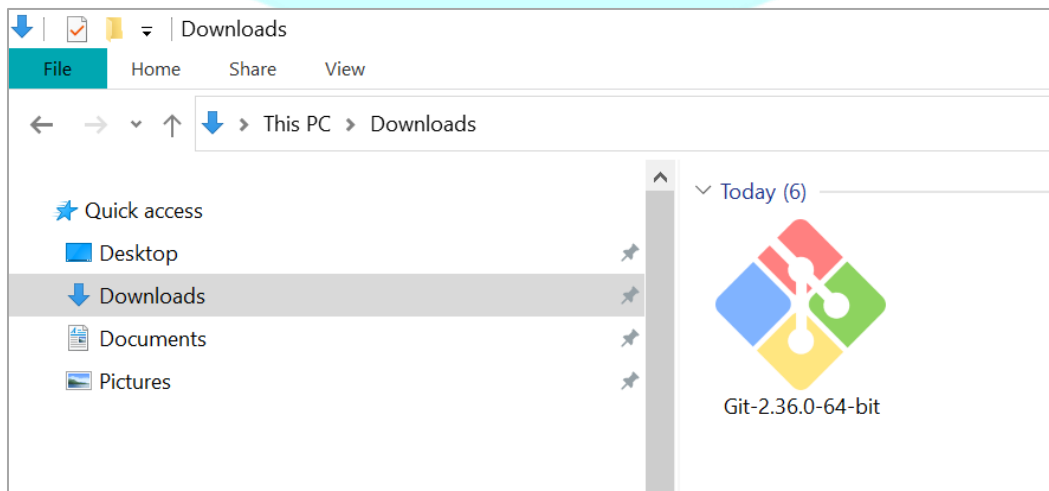
Click the download link for Windows and allow the download to complete.



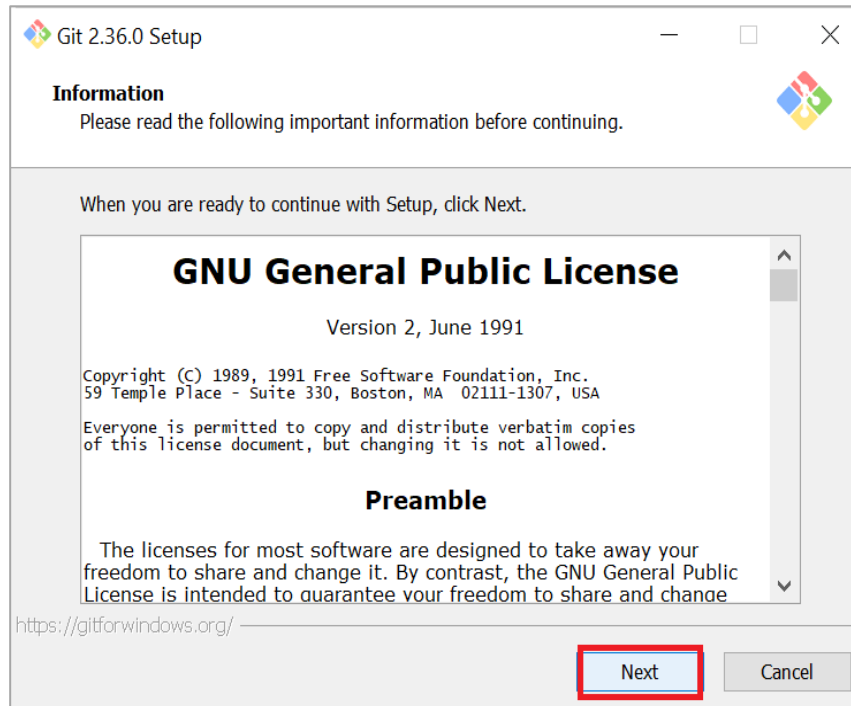
- **Step 2: Extract and Launch Git Installer**

Browse to the download location (or use the download shortcut in your browser).

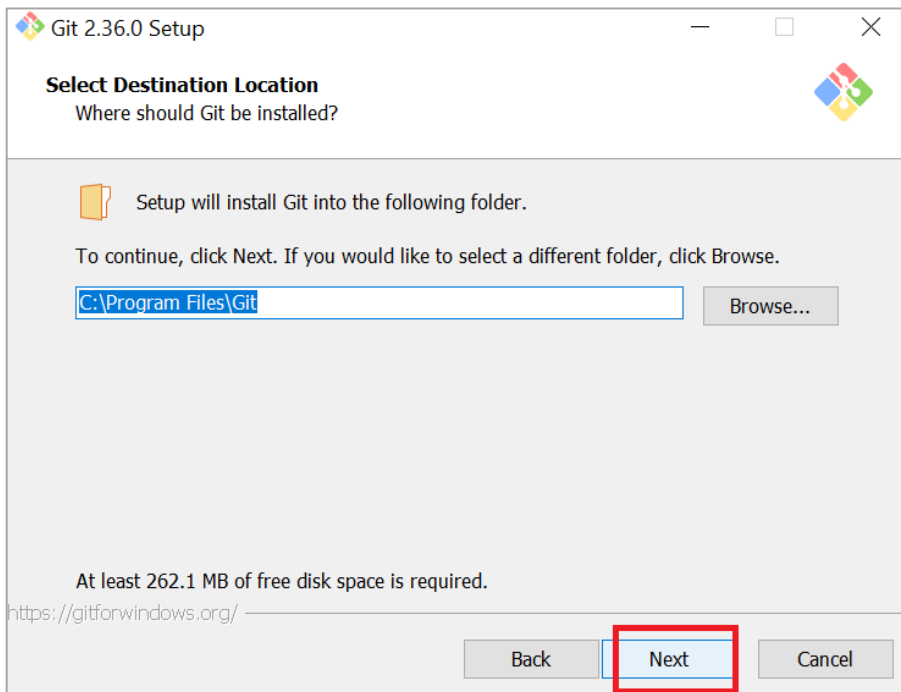
Double-click the file to extract and launch the installer.



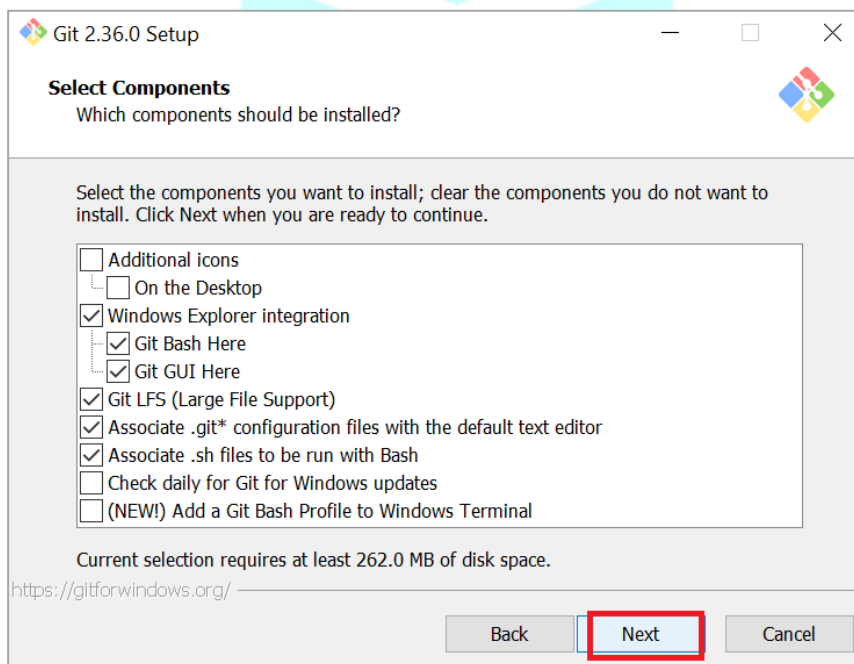
- **Step 3: Allow the app to make changes to your device by clicking Yes on the User Account Control dialog that opens.**
- **Step 4: Review the GNU General Public License, and when you're ready to install, click Next.**



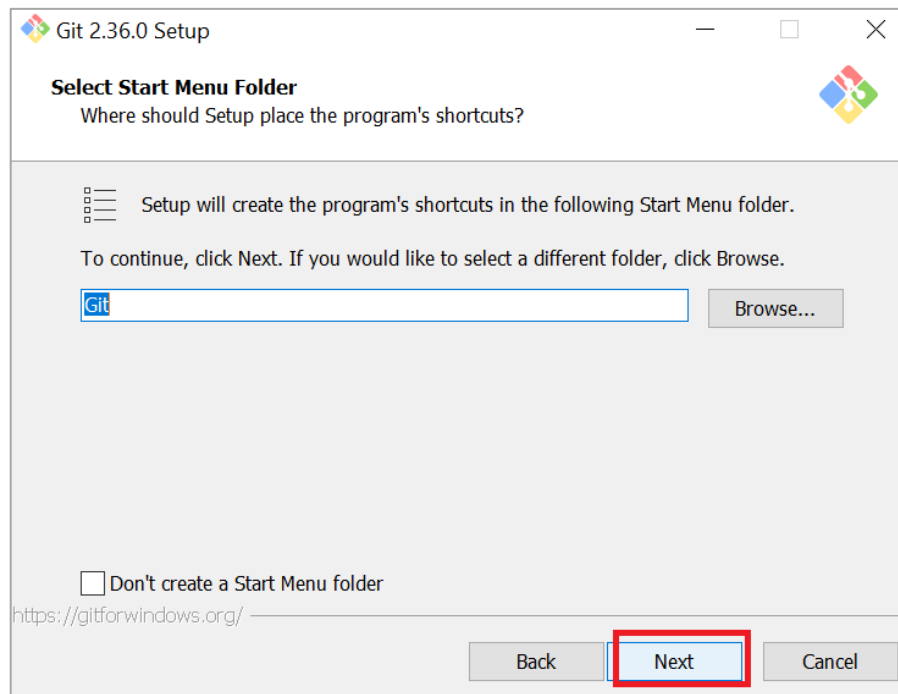
- **Step 5: The installer will ask you for an installation location. Leave the default, unless you have reason to change it, and click Next.**



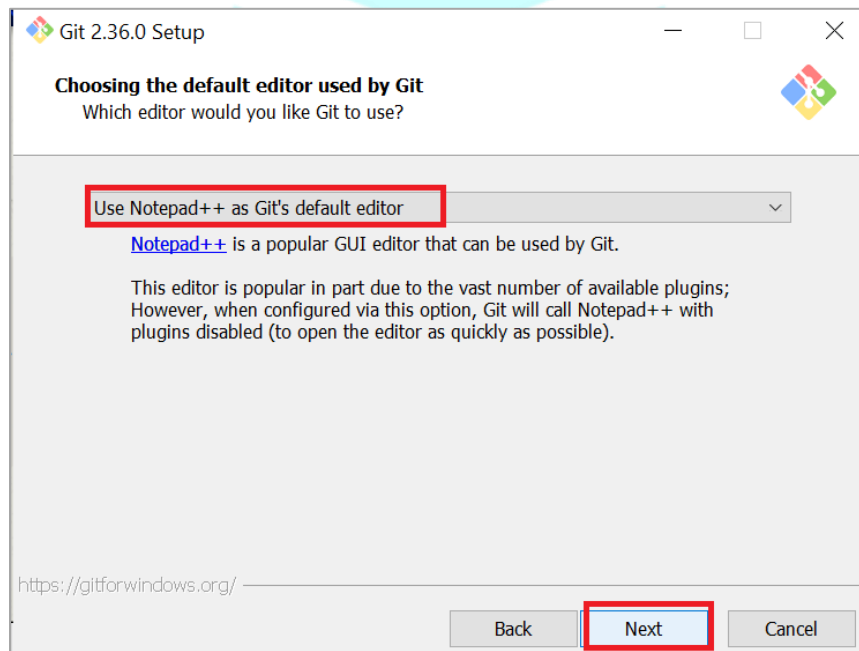
- **Step 6: A component selection screen will appear. Leave the defaults unless you have a specific need to change them and click Next.**



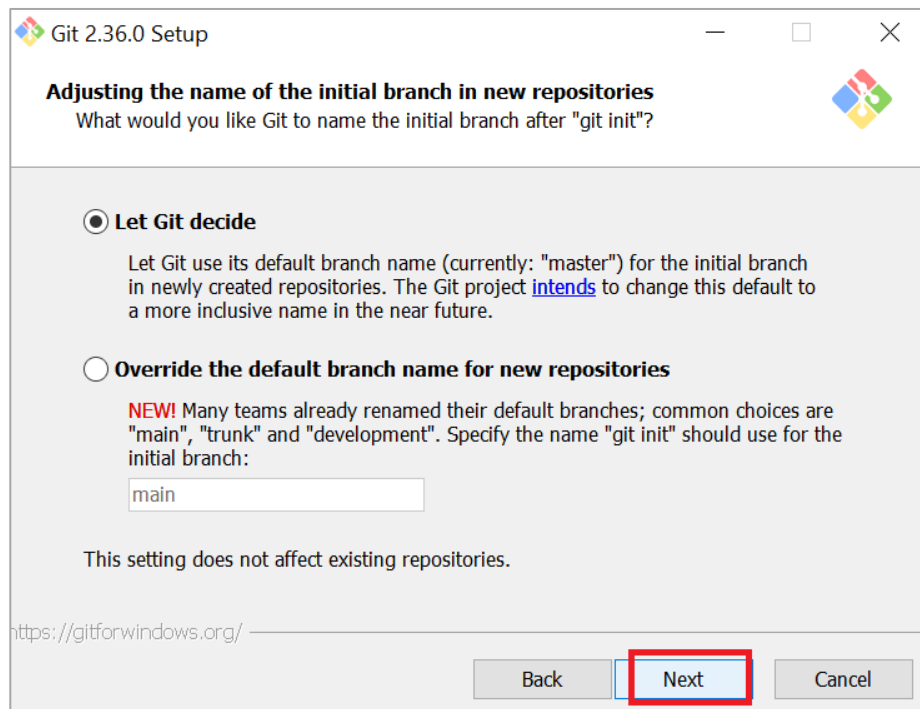
- **Step 7: The installer will offer to create a start menu folder. Simply click Next.**



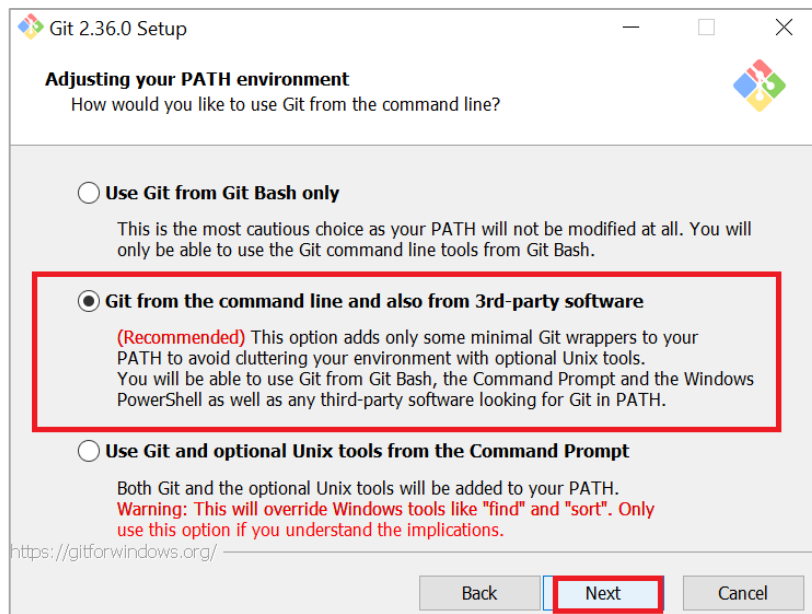
- **Step 8: Select a text editor you'd like to use with Git. Use the drop-down menu to select Notepad++ (or whichever text editor you prefer) and click Next.**



- **Step 9: The next step allows you to choose a different name for your initial branch. The default is 'master.'** Unless you're working in a team that requires a different name, leave the default option and click Next.

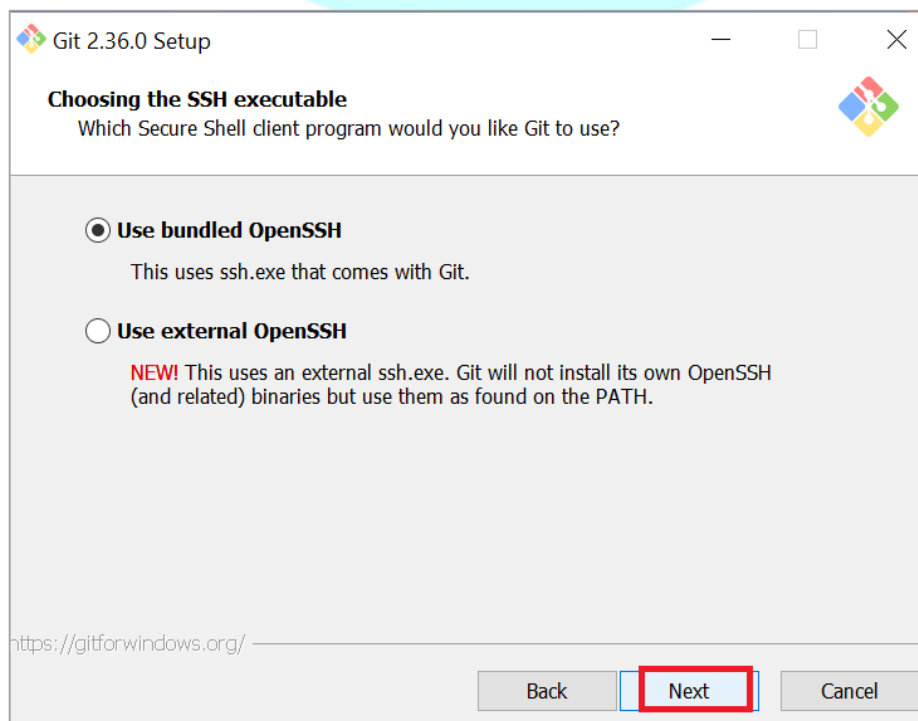


- **Step 10: This installation step allows you to change the PATH environment. The PATH is the default set of directories included when you run a command from the command line. Leave this on the middle (recommended) selection and click Next.**



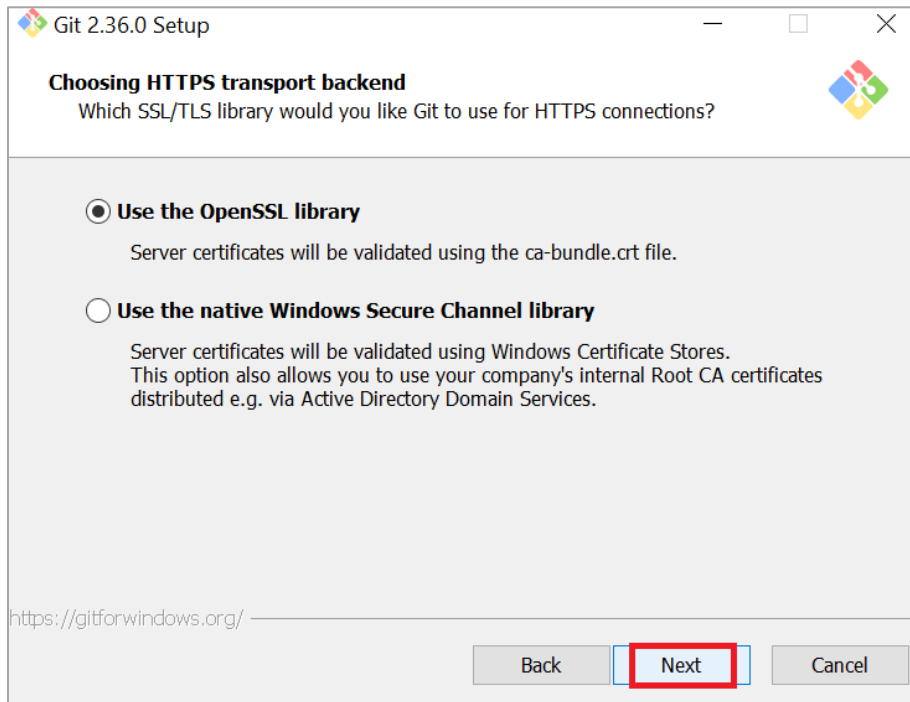
- **Step 11: Server Certificates, Line Endings and Terminal Emulators**

The installer now asks which SSH client you want Git to use. Git already comes with its own SSH client, so if you don't need a specific one, leave the default option and click Next.



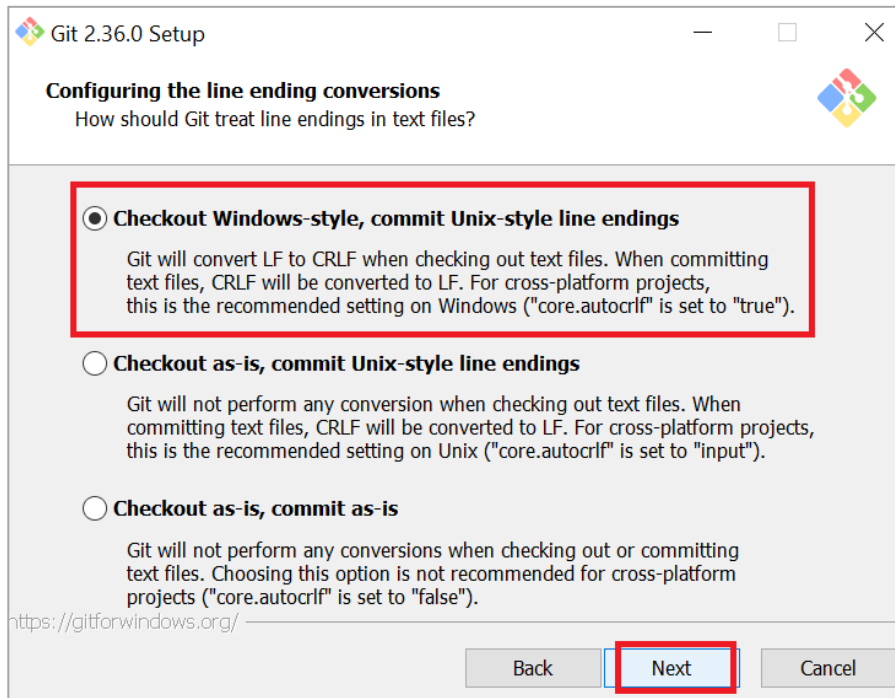
- **Step 12: This option is related to server certificates. Most users should use the default.**

If you're working in an Active Directory Environment, you may need to switch to Windows Store Certificates. Click Next.

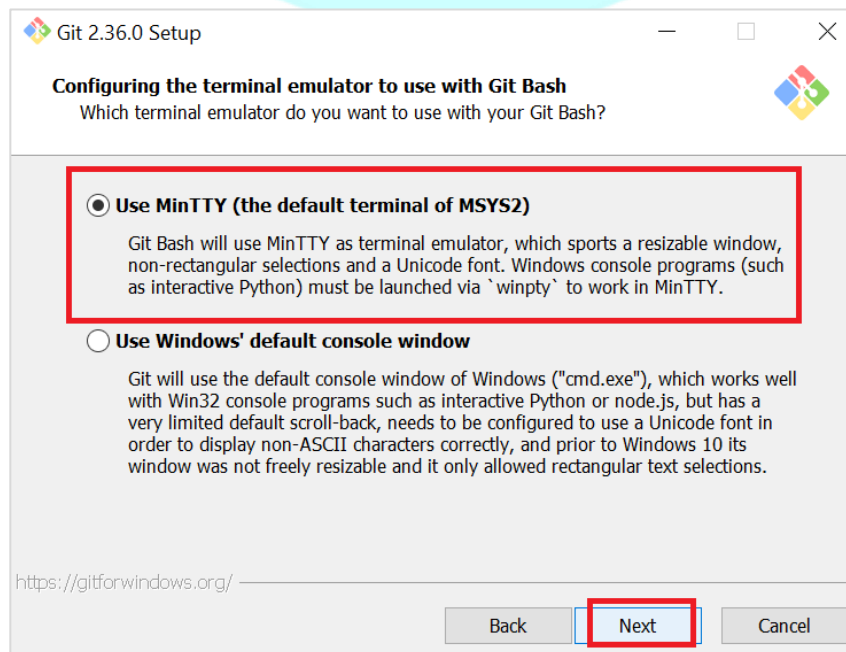


- **Step 13: The next selection converts line endings. It is recommended that you leave the default selection.**

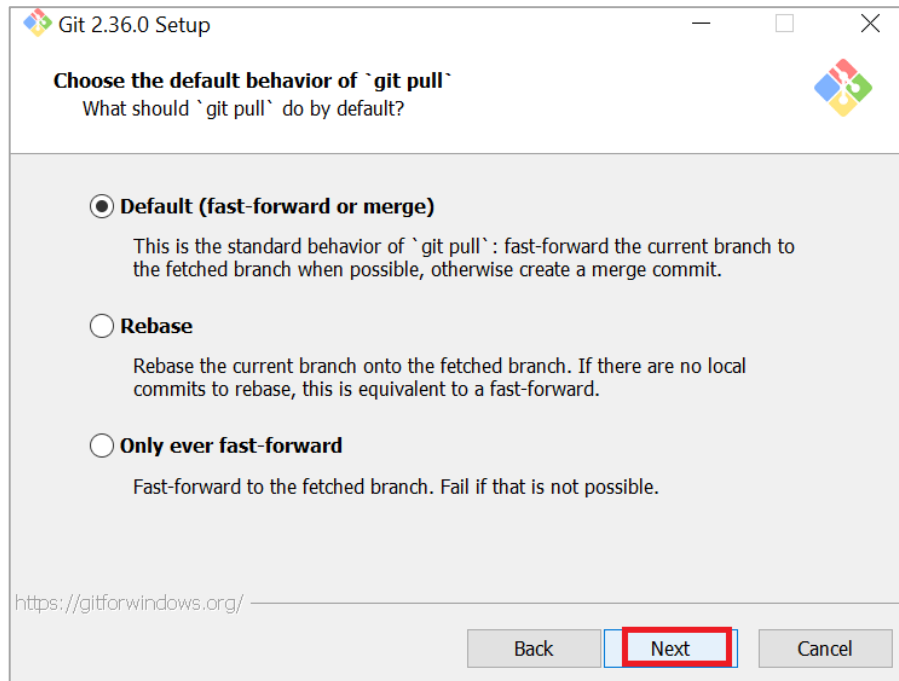
If you're working in an Active Directory Environment, you may need to switch to Windows Store Certificates. Click Next.



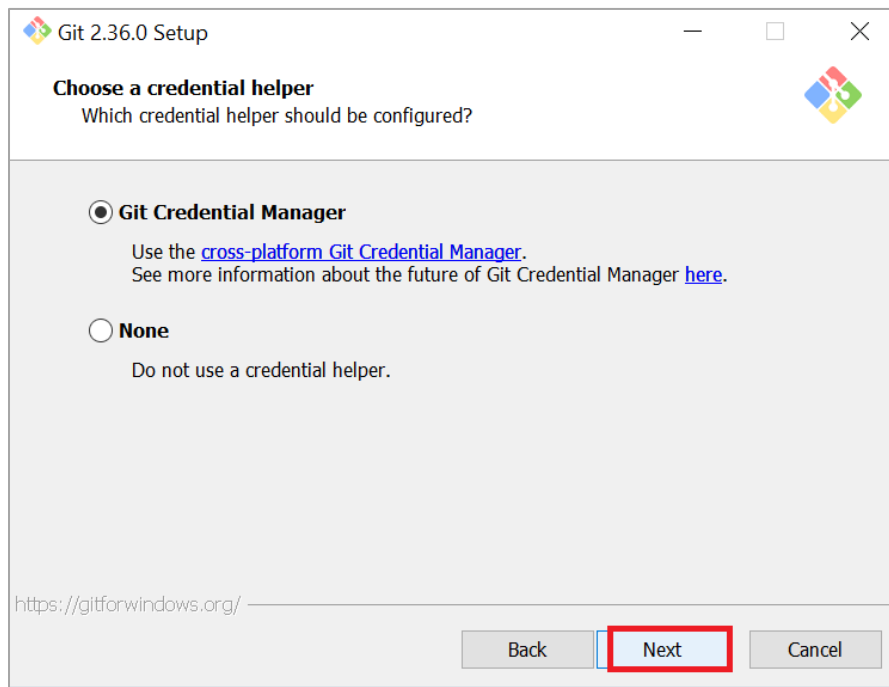
- **Step 14: Choose the terminal emulator you want to use. The default MinTTY is recommended, for its features. Click Next.**



- **Step 15: The installer now asks what the git pull command should do. The default option is recommended unless you specifically need to change its behavior.** Click Next to continue with the installation.



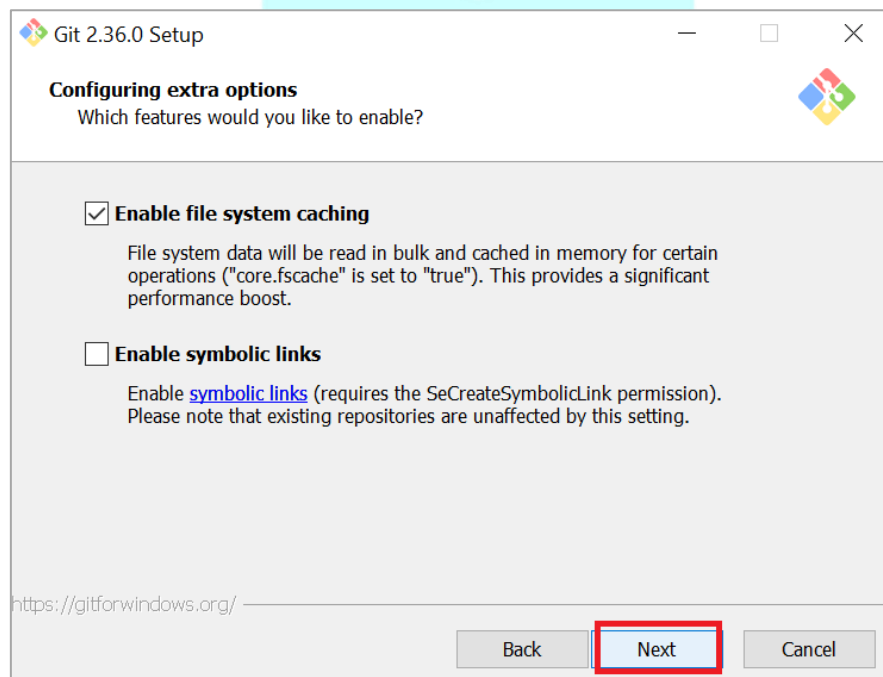
- **Step 16: Next you should choose which credential helper to use. Git uses credential helpers to fetch or save credentials.** Leave the default option as it is the most stable one, and click Next.



- **Step 17: Additional Customization Options**

The default options are recommended;

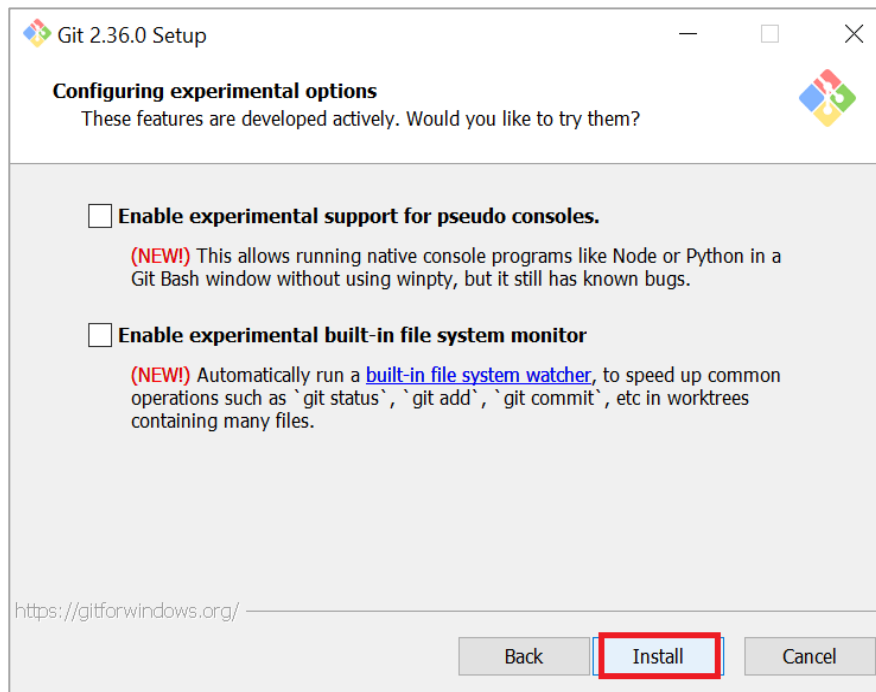
However this step allows you to decide which extra option you would like to enable. If you use symbolic links, which are like shortcuts for the command line, tick the box. Click **Next**.



- **Step 18: Depending on the version of Git you're installing; it may offer to install experimental features.**

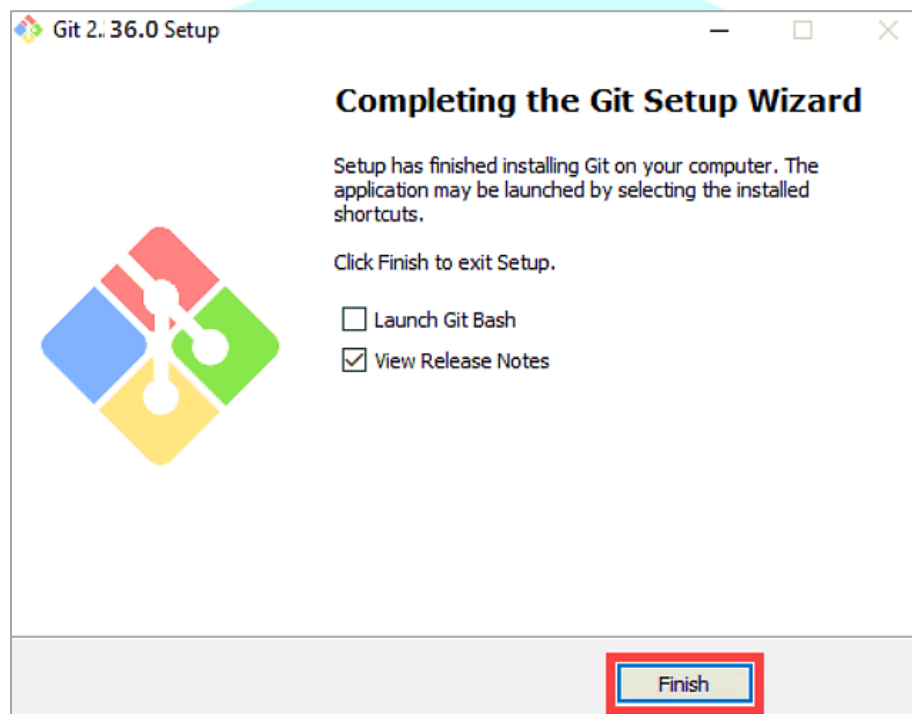
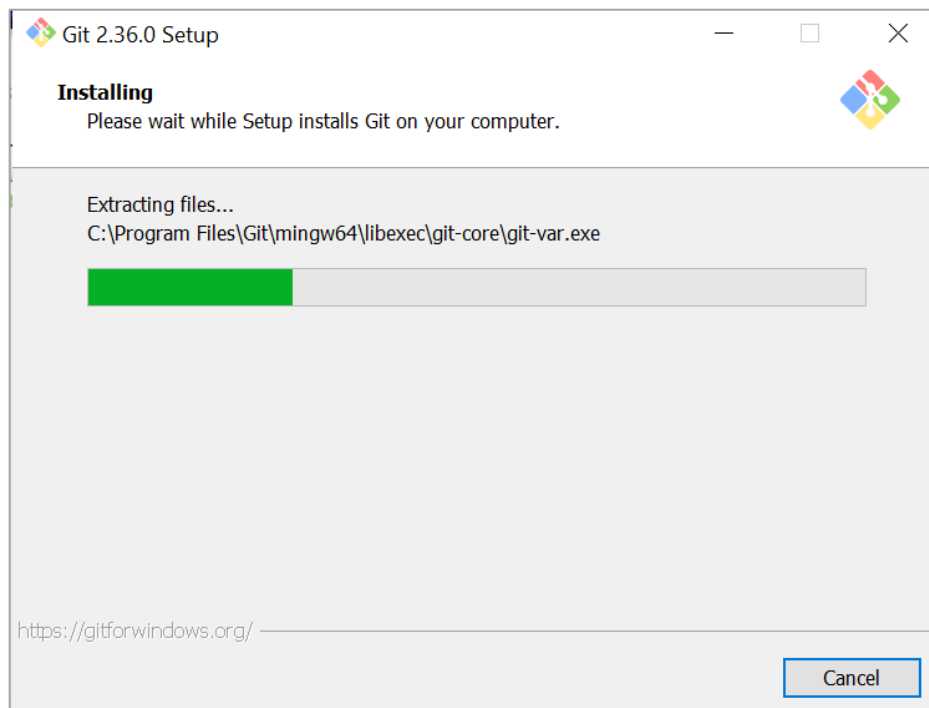
At the time this article was written, the options to include support for pseudo controls and a built-in file system monitor were offered.

Unless you are feeling adventurous, leave them unchecked and click Install.



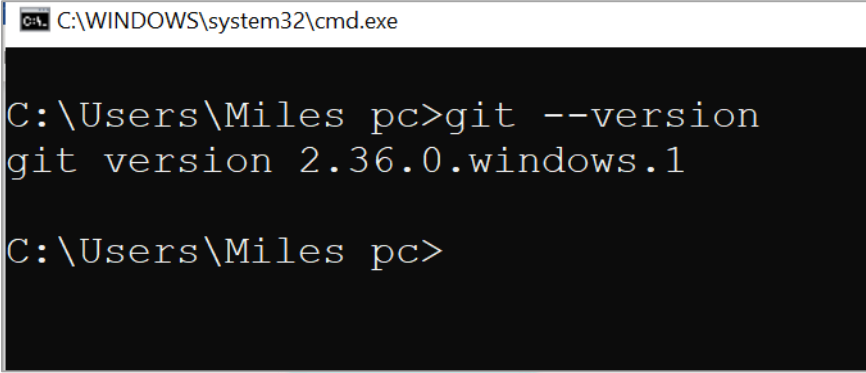
- **Step 19: Complete Git Installation Process**

Once the installation is complete, tick the boxes to view the Release Notes or Launch Git Bash, then click **Finish**.



- **Step 20: Verify the Git version**

To check if Git has successfully installed on your system, run **git --version** command on the command prompt.



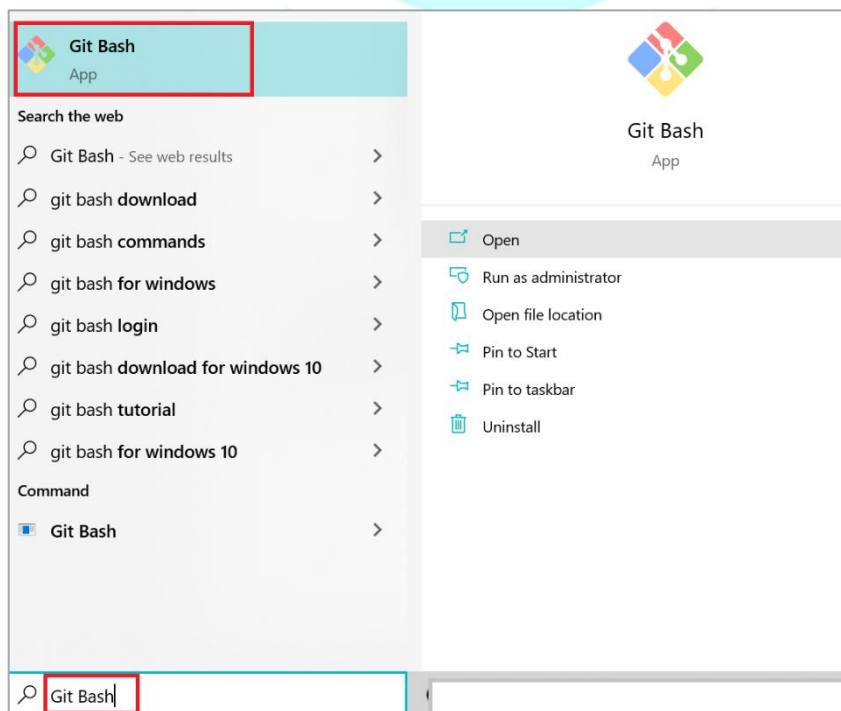
```
C:\WINDOWS\system32\cmd.exe

C:\Users\Miles pc>git --version
git version 2.36.0.windows.1

C:\Users\Miles pc>
```

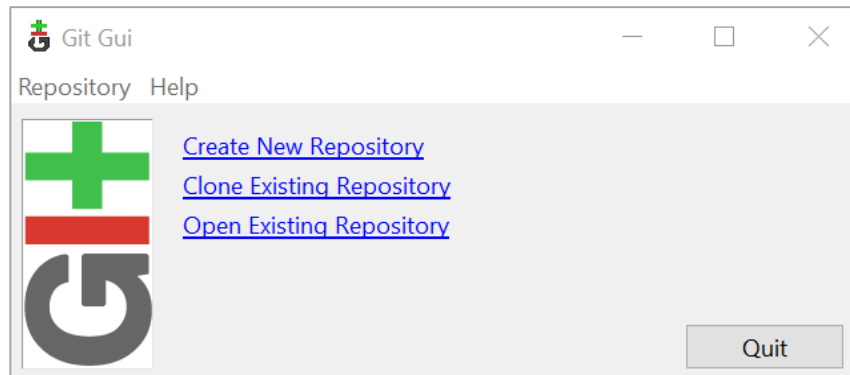
- **Step 21: Launch Git Bash Shell**

To launch **Git Bash** open the **Windows Start** menu, type *git bash* , and press **Enter** (or click the application icon).



- **Step 22: Launch Git GUI**

To launch **Git GUI** open the **Windows Start** menu, type **git gui** and press **Enter** (or click the application icon).



You're all set!

4. Node.js Installation

Node.js is a JavaScript runtime environment that allows you to execute code outside of a browser. node.js also provides a library of modules that makes it easy to create web applications.

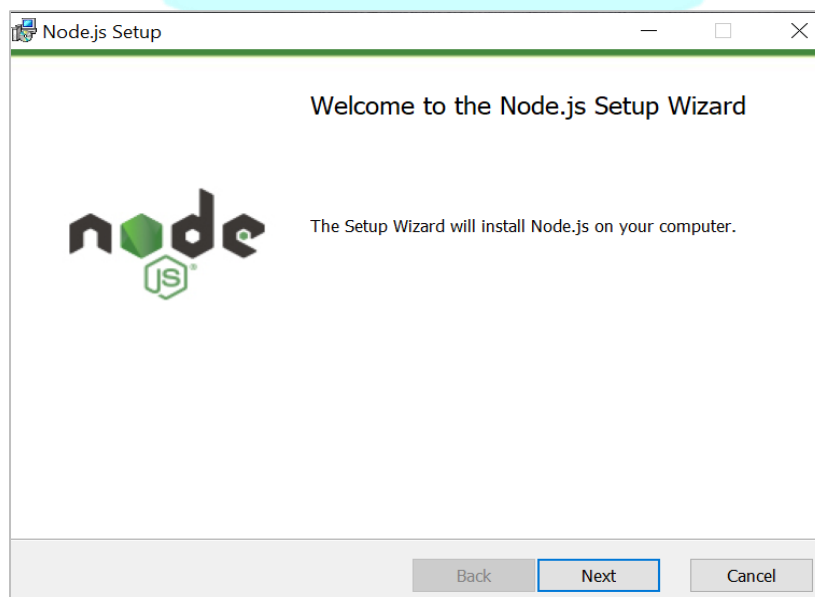
- **Step 1: Install Node.js**

Follow the link: <https://nodejs.org/en/download/> Download the Node.js installer for Windows and install it.

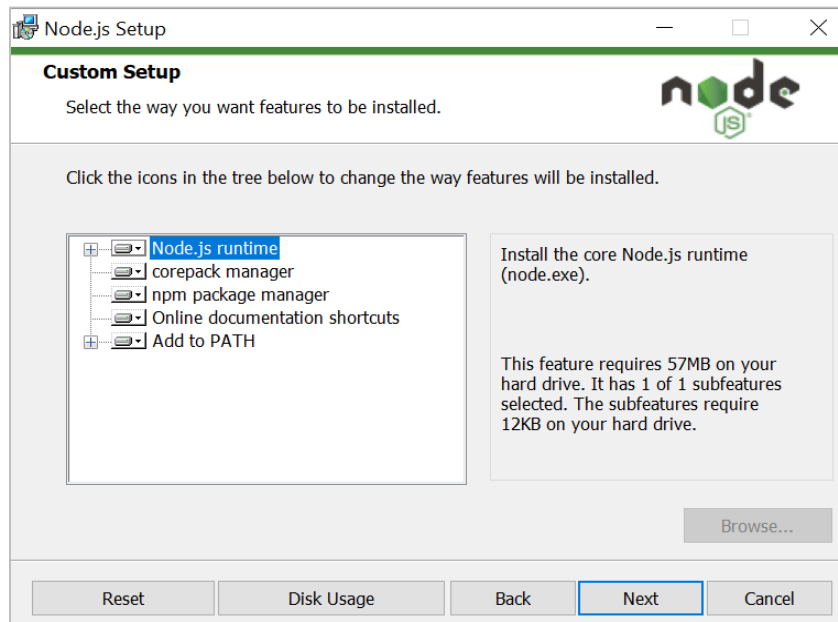
LTS Recommended For Most Users	Current Latest Features															
 Windows Installer node-v16.15.0-x64.msi	 macOS Installer node-v16.15.0.pkg	 Source Code node-v16.15.0.tar.gz														
Windows Installer (.msi) Windows Binary (.zip) macOS Installer (.pkg) macOS Binary (.tar.gz) Linux Binaries (x64) Linux Binaries (ARM) Source Code	<table border="1"> <tr> <td>32-bit</td> <td>64-bit</td> </tr> <tr> <td>32-bit</td> <td>64-bit</td> </tr> <tr> <td colspan="2">64-bit / ARM64</td> </tr> <tr> <td>64-bit</td> <td>ARM64</td> </tr> <tr> <td colspan="2">64-bit</td> </tr> <tr> <td>ARMv7</td> <td>ARMv8</td> </tr> <tr> <td colspan="2">node-v16.15.0.tar.gz</td> </tr> </table>		32-bit	64-bit	32-bit	64-bit	64-bit / ARM64		64-bit	ARM64	64-bit		ARMv7	ARMv8	node-v16.15.0.tar.gz	
32-bit	64-bit															
32-bit	64-bit															
64-bit / ARM64																
64-bit	ARM64															
64-bit																
ARMv7	ARMv8															
node-v16.15.0.tar.gz																

- **Step 2: Open Setup File**

Now from Setup file, Start the Installation Wizard.



- **Step 3: Click Agree and Make sure all the options are installed.**



- **Step 4: Finish the installation and open Command prompt/Terminal.**

To verify the installation, Type `node --version` and `npm --version` to check the versions of Node and NPM.

A screenshot of a Windows Command Prompt window. The title bar says 'Command Prompt'. The prompt is at 'C:\Users\Miles pc>'. The first command entered is 'node --version', and the output is 'v16.15.0'. The second command entered is 'npm --version', and the output is '8.5.5'. The prompt is now at 'C:\Users\Miles pc>' with a cursor.

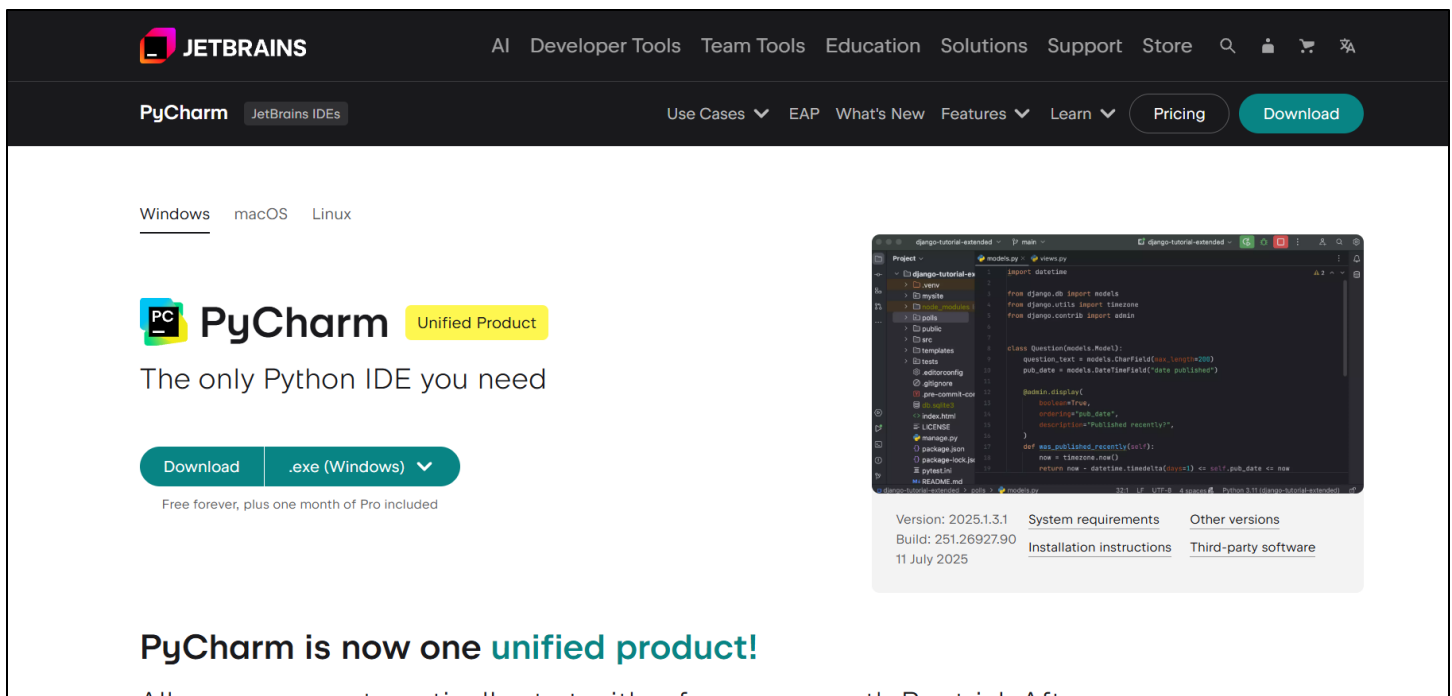
5. PyCharm IDE Installation

PyCharm is a powerful IDE used for Python development. It offers excellent code analysis, graphical debugger, integrated testing, and version control system support.

Installing PyCharm is simple. Follow the steps below to get started.

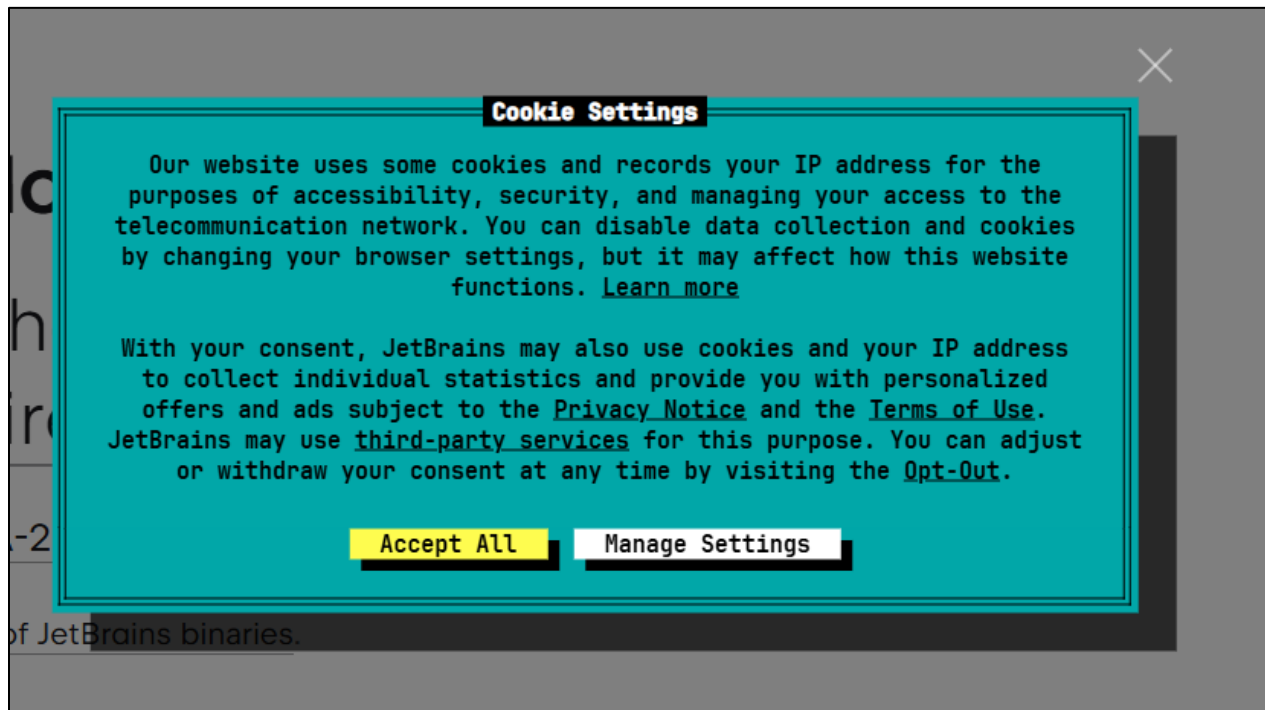
- **Step 1: Download from Official Website:**

<https://www.jetbrains.com/pycharm/download/?section=windows>

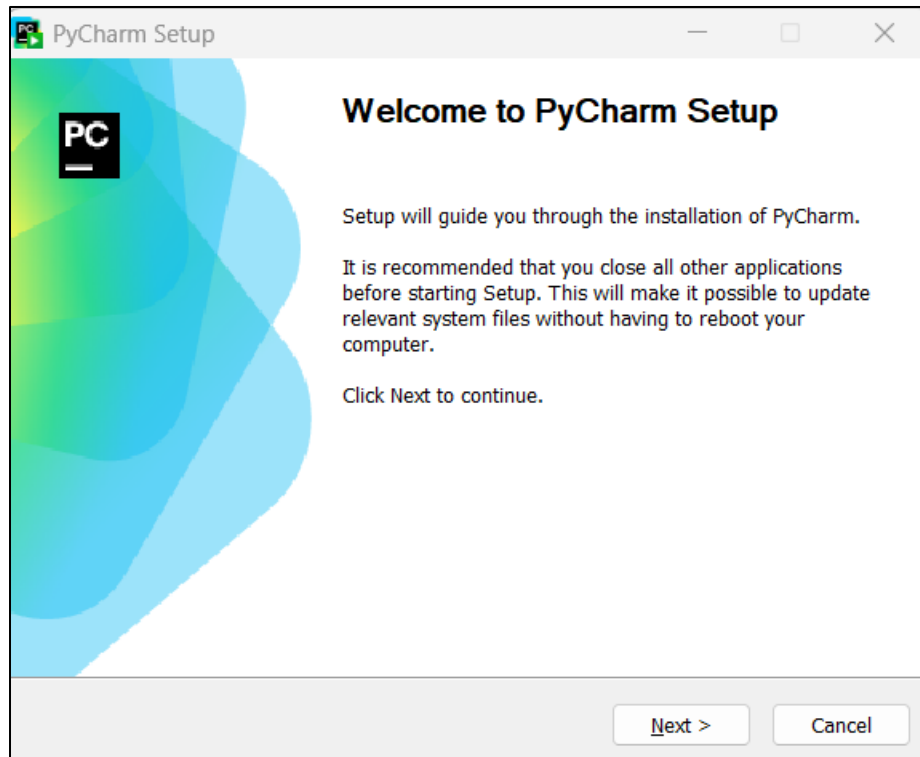


The screenshot shows the JetBrains website's PyCharm download page for Windows. The page features the JetBrains logo, navigation links for AI, Developer Tools, Team Tools, Education, Solutions, Support, Store, and a search icon. Below the navigation bar, there are tabs for 'PyCharm' and 'JetBrains IDEs', and a 'Download' button. The main content area has tabs for 'Windows', 'macOS', and 'Linux'. The 'Windows' tab is active, showing the PyCharm logo, the text 'Unified Product', and 'The only Python IDE you need'. A 'Download' button is present, with a dropdown menu showing '.exe (Windows)'. Below the button, it says 'Free forever, plus one month of Pro included'. To the right, there is a preview of the PyCharm IDE interface showing a Django project. Below the preview, there are links for 'System requirements', 'Installation instructions', 'Other versions', and 'Third-party software'. At the bottom, it says 'PyCharm is now one unified product!' and 'All users now automatically start with a free one-month Pro trial. After'.

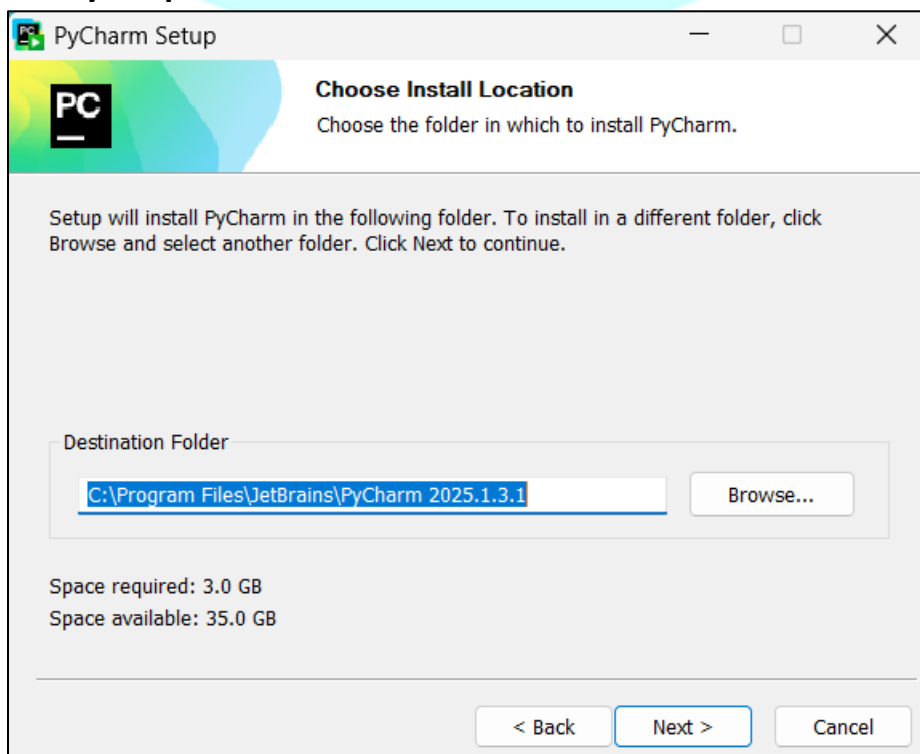
- **Step 2: Navigate to your download location. Double-click the downloaded .exe file to launch the PyCharm Setup Wizard.**
- **Step 3: Cookie Consent: Choose “Accept All” to allow cookies or “Manage Settings” to customize preferences.**



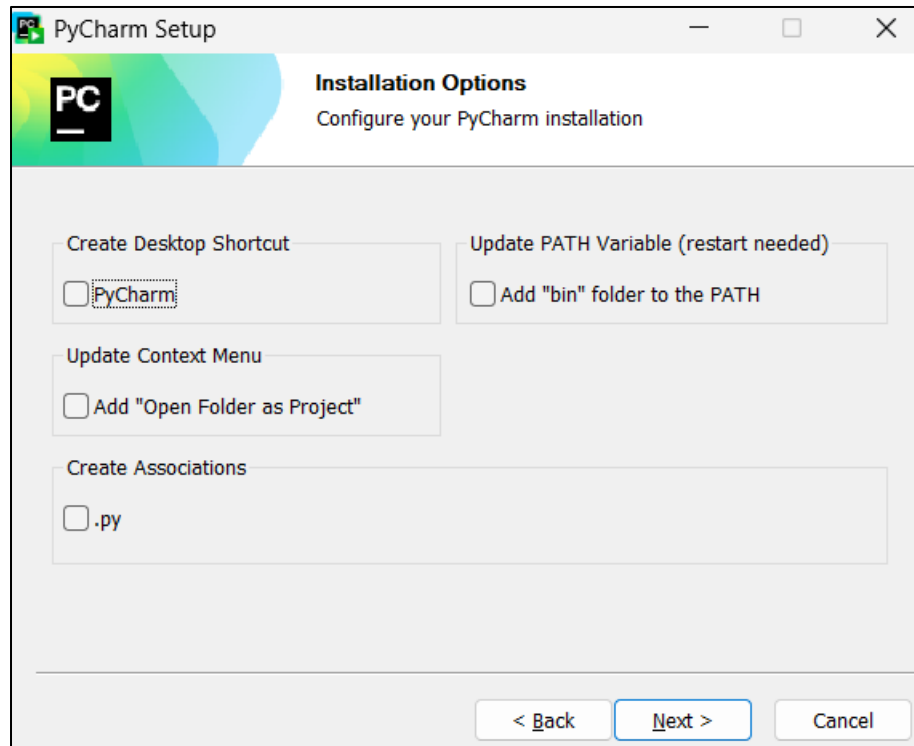
- **Step 4: Click on "Download" and then click "Next."**



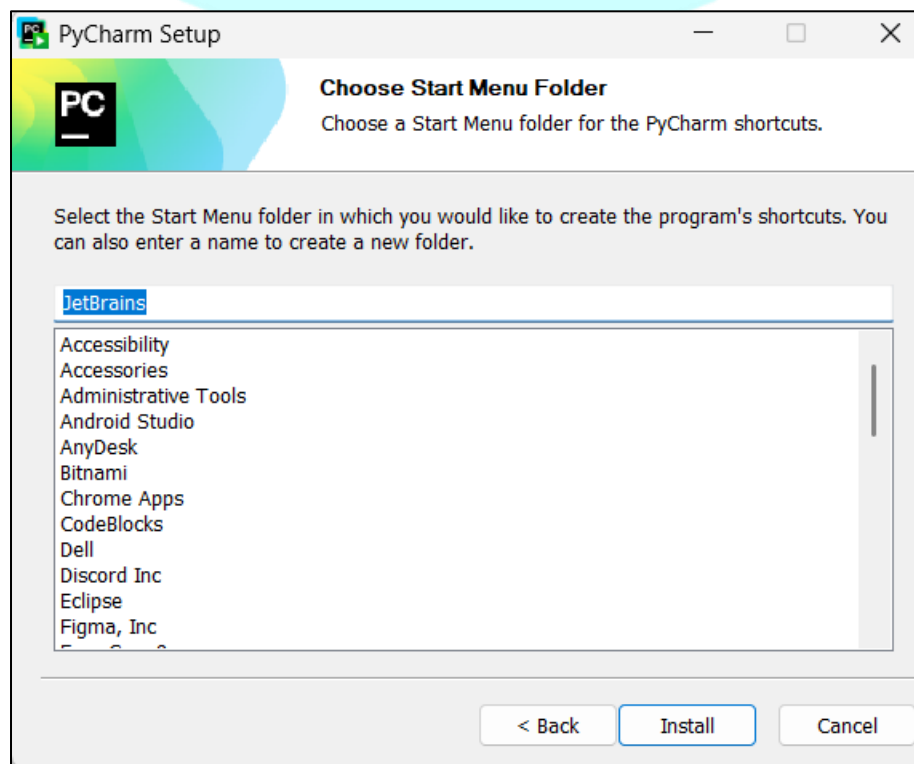
- **Step 5: Choose your preferred destination folder for installation.**



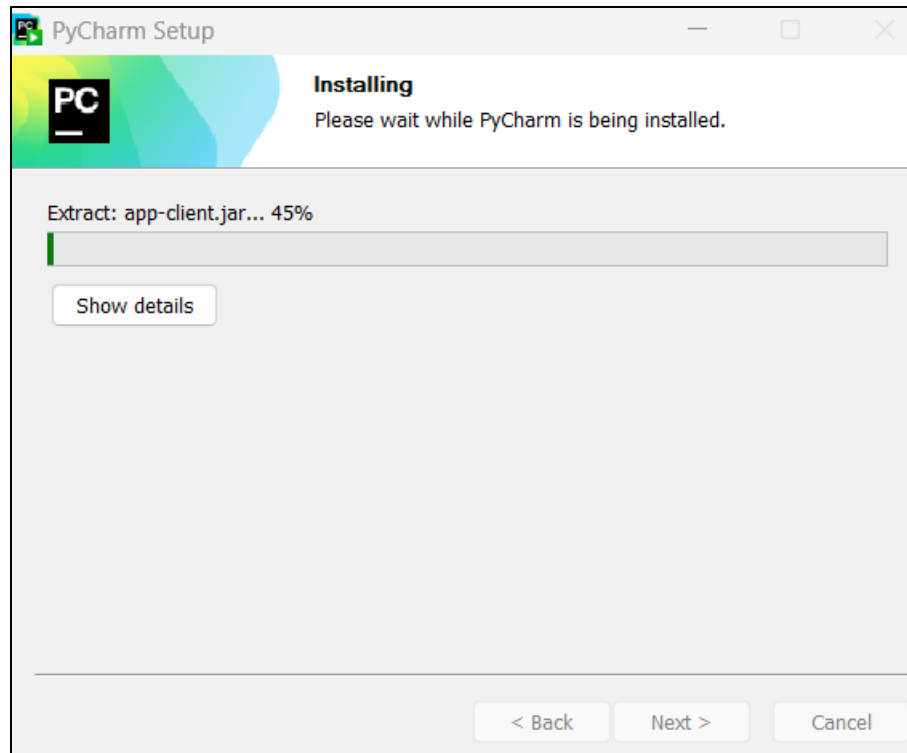
- **Step 6: Select installation options according to your preferences.**



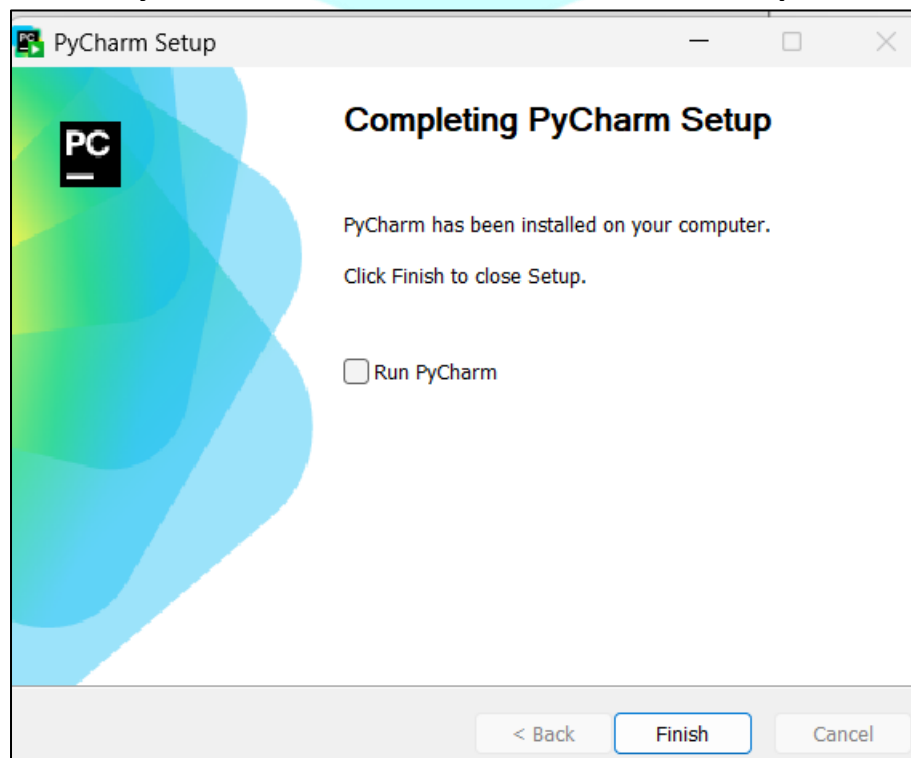
- **Step 7: Select JetBrains and click "Install."**



- **Step 8: Allow the installation process to complete.**

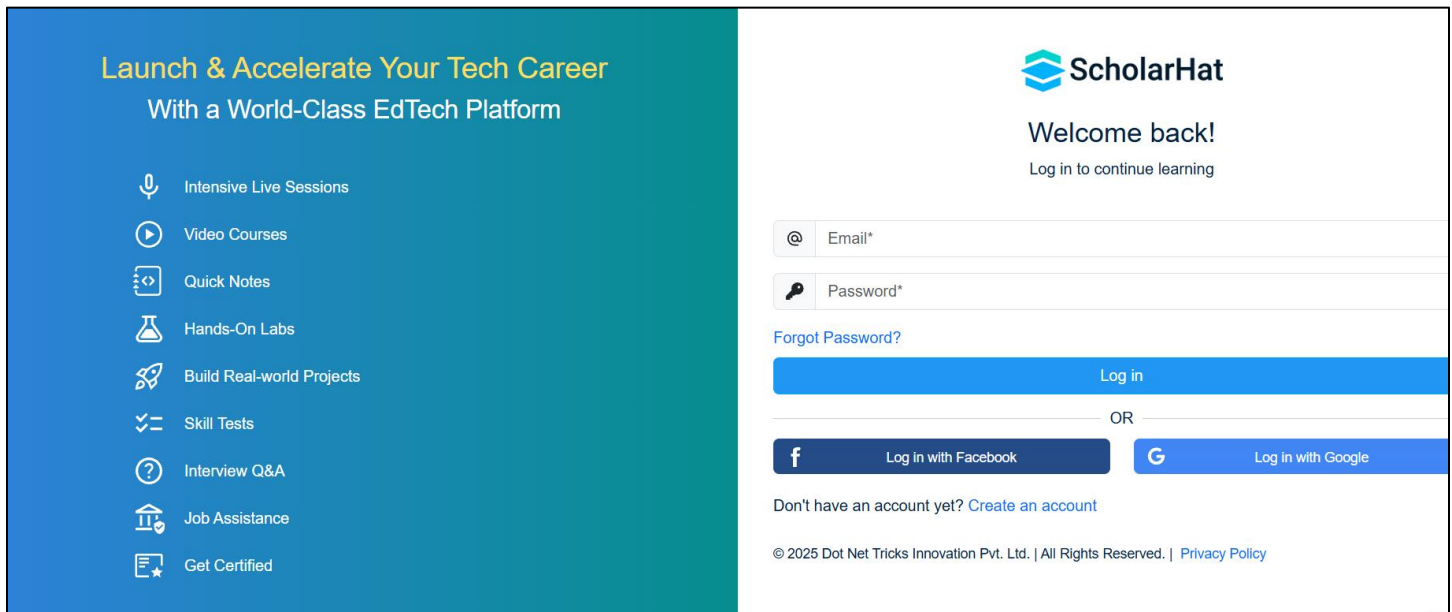


- **Step 9: Once installation is finished, PyCharm will indicate successful installation. Click "I want to manually reboot later," then click "Finish" to complete the process.**

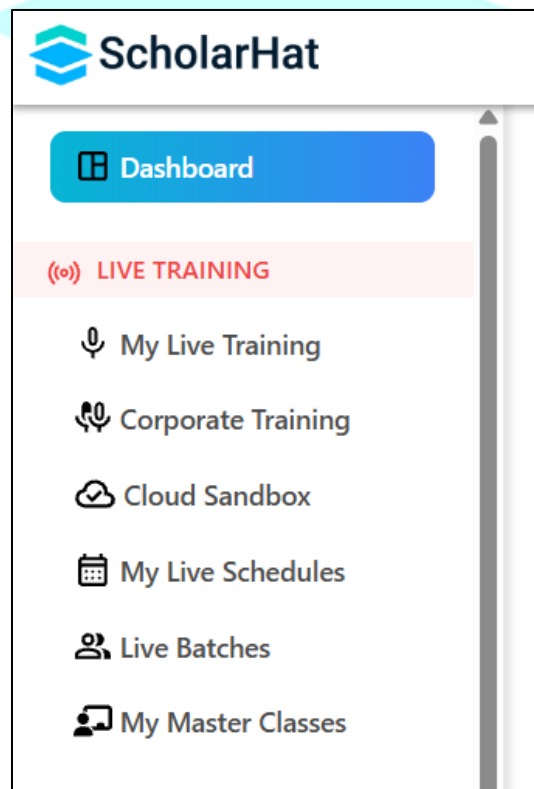


6. Azure AI Cloud Sandbox

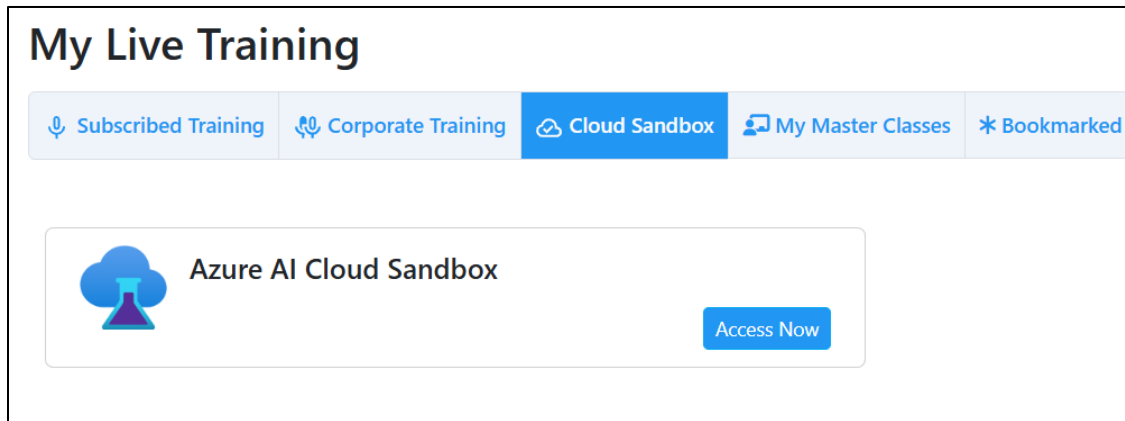
- **Step 1: Visit the ScholarHat website and log in with your username and password.**



- **Step 2: Navigate to the “Cloud Sandbox” tab from the dashboard after logging in.**



- **Step 3: Under “Azure AI Cloud Sandbox”, click on the “Access Now” button to proceed.**



- **Step 4: Click “Start Cloud Sandbox” to launch your Azure environment.**



Azure AI Cloud Sandbox

Expires on: Fri, Oct 17, 2025

Sandbox duration 1 hour

Average Start time Less than a minute

[Start Cloud Sandbox](#)

How to Access Cloud Sandbox - Azure AI

Step1: Sign in to Azure Portal

- Go to the Azure portal by clicking on the Open Console button or by using URL <https://portal.azure.com>.

Tips:

- Use Incognito/Private mode to avoid browser cache issues with Azure Portal.
- If you are logged into a different Azure account, please sign out and clear browser cache before proceeding.

Step2: Use Your Provided Credentials

- Enter the email address and password provided to you for the Cloud Sandbox.

Need Help?

-  How to use cloud sandbox
-  Troubleshooting Sandbox

[Submit Feedback](#)

- **Step 5: Confirm by clicking “Start Sandbox” on the popup. Make sure the terms checkbox is selected.**

Confirmation

Are you sure you want to start Sandbox?

☒ By participating in cloud sandbox, I agree to be bound by Cloud Sandbox Terms of Use .

Start SandboxCancel

- **Step 6: Once the sandbox starts, copy the email and password under Sandbox Credentials for logging into the Azure portal.**

00h 59m 17s left

End Sandbox

Open Console

SANDBOX CREDENTIALS

Email:

pragatikankar_1655667@scholarhatlab.onmicrosoft.com



Password:

K5BDMH4EDKQJ0MMS

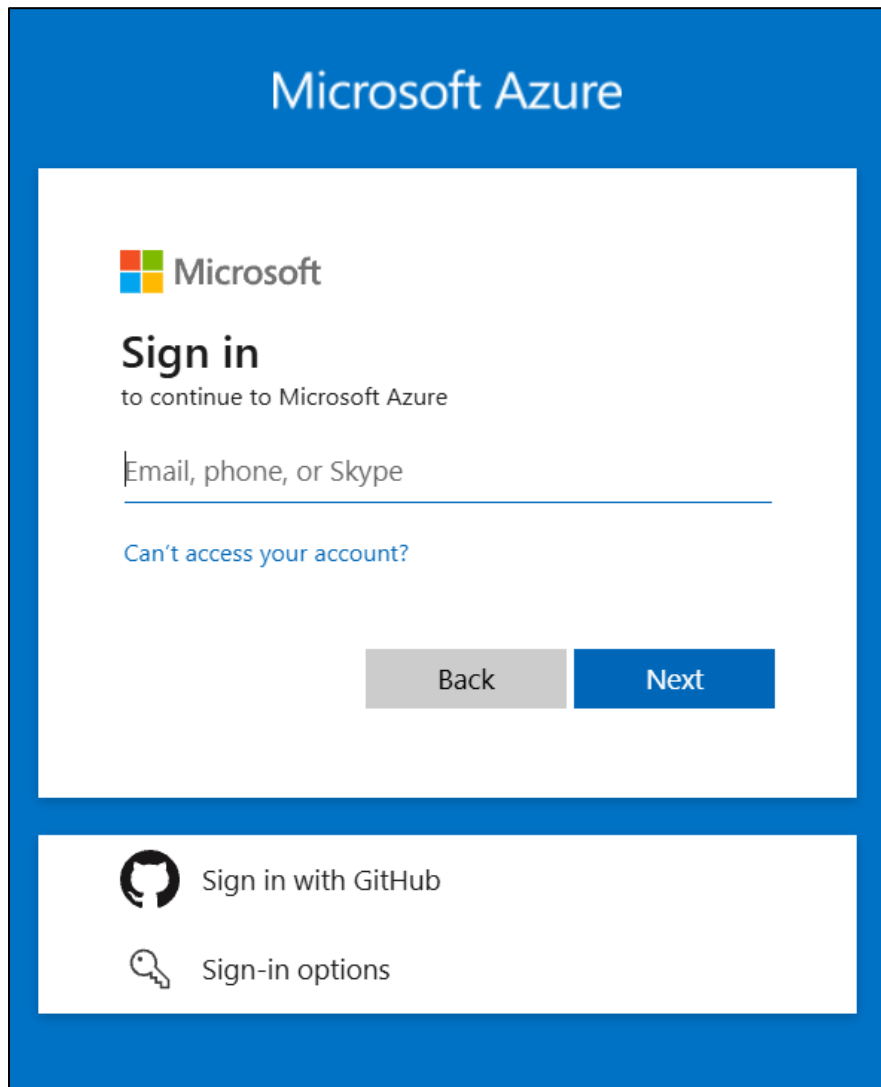


Resource Group:

scholar-rg-4653

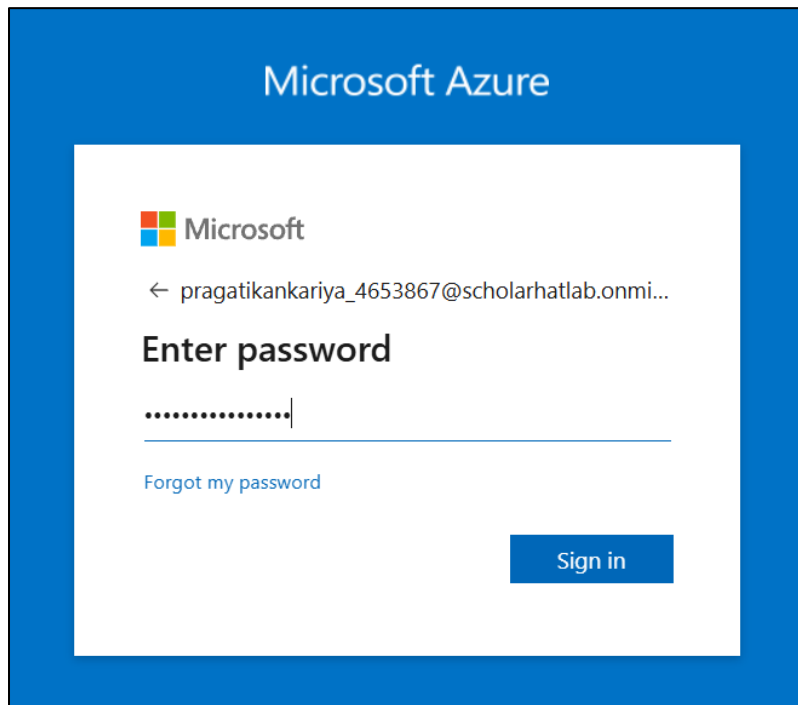


- **Step 7: Sign in to Azure Portal:** Go to the Azure portal by clicking “Open Console” in ScholarHat or visiting <https://portal.azure.com>.

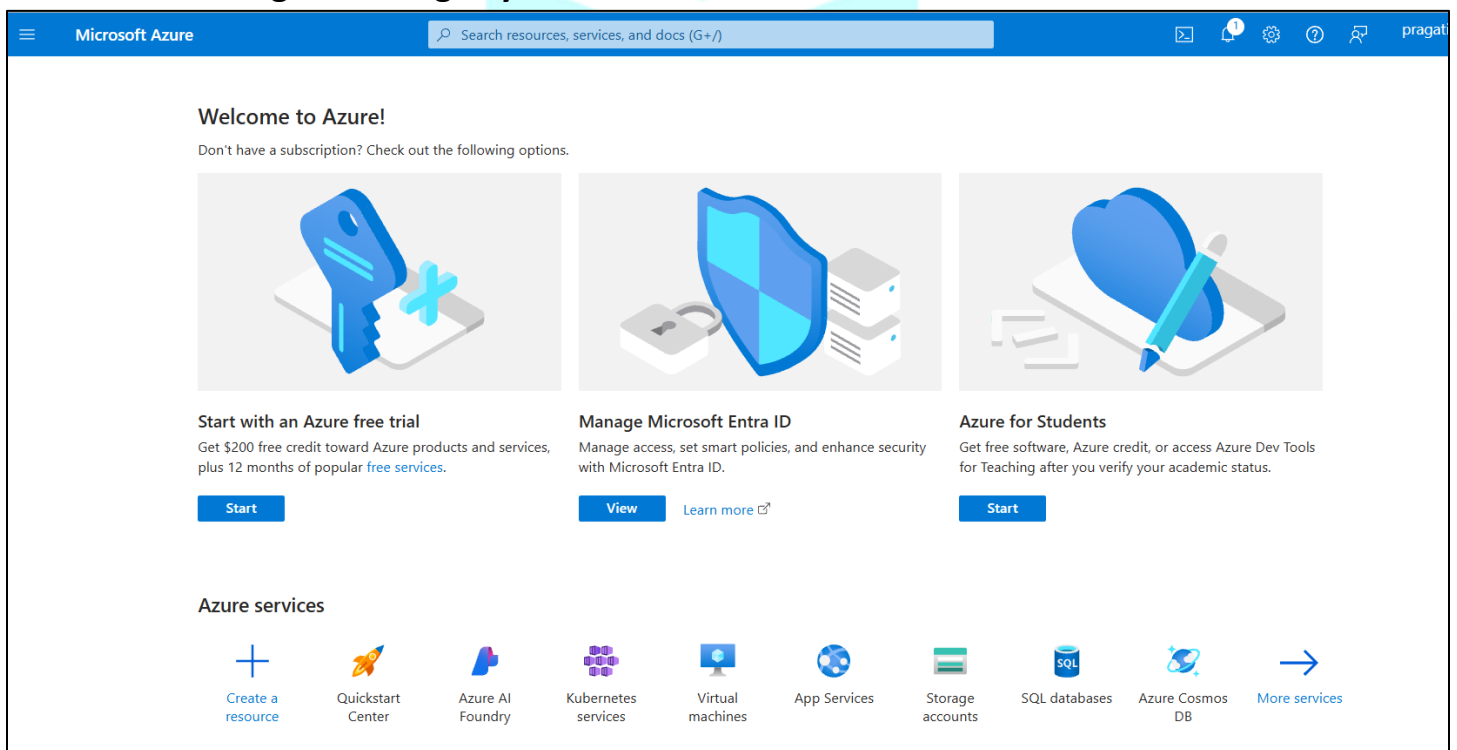


The image shows the Microsoft Azure sign-in interface. At the top, the text "Microsoft Azure" is displayed in white on a blue background. Below this, the Microsoft logo is followed by the text "Sign in" and "to continue to Microsoft Azure". A text input field is labeled "Email, phone, or Skype". Below the input field is a link that says "Can't access your account?". At the bottom of the sign-in section are two buttons: "Back" (disabled, grey) and "Next" (active, blue). Below the sign-in section, there are two options: "Sign in with GitHub" with the GitHub logo and "Sign-in options" with a key icon.

- **Step 8: Use Your Provided Credentials: Enter the email ID and password provided to you for Cloud Sandbox access and click Next.**



- **Step 7: After logging into the Azure portal, you'll land on the Azure Dashboard. From here, you can explore services like Virtual Machines, App Services, SQL Databases, and more to begin working in your sandbox environment.**



Contact Us

If you are still facing any issues while installing the software and configuring the system, please free to reach out to us at **+91 9999 123 502** or email us at hello@scholarhat.com

